


Degree Project in Innovation Management and Product Development

Second Cycle 30 Credits

# Visions and Challenges in Product-Service System Change

An Exploratory Study of Aftersales Systems for Driverless Trucks based on Industry Practices

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Master of Science Thesis TRITA-ITM-EX 2024:232
KTH Industrial Engineering and Management
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SE-100 44 STOCKHOLM

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**Examensarbete TRITA-ITM-EX 2024:232**

# Visioner och utmaningar i förändring av produktservicesystem: En explorativ studie av eftermarknadssystem för förarlösa lastbilar baserad på branschpraxis

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# *SAMMANFATTNING*

Inom lastbilsindustrin pågår ett skifte till förarlösa lastbilar. Det finns dock begränsad forskning om eftermarknadsaspekterna av förarlösa lastbilar. Syftet med detta examensarbete är därför att undersöka förväntade utmaningar och nuvarande visioner för eftermarknadssystemet för förarlösa lastbilar inom Scania CV och Traton. Studien syftar till att identifiera potentiella hinder i utvecklingen av eftermarknaden för förarlösa lastbilar i linjetransport och föreslår riktlinjer för att hantera utvecklingen. Linjetransporter är defienrat som längre transporter mellan terminaler.

Examensarbetet täcker flera teoretiska områden som är relevanta för ämnet, inklusive produktservicesystem, ambidextriska organisationer, förändringsledning, visioner inom organisationer och hantering av tvetydigheter. Studien genomfördes genom två fältbesök på Scania verkstäder för att skapa en förståelse för den nuvarande utformningen av eftermarknadssystemet, diskussioner med två experter inom Scania CV och två intervjuomgångar med totalt 14 respondenter från relevanta avdelningar inom Scania CV och Traton. Intervjuerna utfördes för att samla in empiriska data om de nuvarande visionerna för ett eftermarknadssystem för förarlösa lastbilar. Den första intervjuomgången var mestadels med högre chefer och den andra omgången med lägre chefer.

Enligt resultaten finns det olika tvetydigheter och utmaningar när det gäller både tekniska och affärsrelaterade aspekter av utformningen av eftermarknadssystemet för förarlösa lastbilstillämpningar, vilket skapar ett behov av samarbete mellan de berörda avdelningarna. Det finns olika visioner mellan avdelningarna och inom dem. Olika teorier kan stödja samarbetsprocessen och hantera tvetydighet och med hjälp av den empiriska datan kommer en analys av den nuvarande samarbetsstrukturen presenteras. I kombination med detta kommer även förslag presentears gällande hur mer gynnsamma förutsättningar för eftermarknadsutveckling kan etableras.

**Nyckelord:** Förarlösa lastbilar, Eftermarknad, Produktservicesystem, Ambidextriska organisationer, Förändringsledning, Visioner, Tvetydighet.

KTH Industrial Engineering and Management

**Master of Science Thesis TRITA-ITM-EX 2024:232**

# Vision and Challenges in Product-Service System Change: An Explorative Study of Aftersales Systems for Driverless Trucks based on Industry practices

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# *ABSTRACT*

There is an ongoing transition towards driverless trucks within the truck industry. However, there is limited research regarding the aftersales aspects of driverless trucks. Therefore, the purpose of this thesis project is to investigate expected challenges and current visions of the aftersales system for driverless trucks within the organization of Scania CV and Traton. The study aims to identify potential obstacles in the development of aftersales activities for driverless trucks in hub-to-hub contexts and proposes guidelines for managing the development. Hub-to-hub is defined as long-haulage transports between terminals.

The thesis project covers several theoretical fields relevant to the subject, including product service systems, ambidextrous organizations, change management, visions within organizations, and handling ambiguity. The study was conducted by executing two field visits at Scania workshops to establish an understanding of the current aftersales system design, discussions with two experts within Scania CV, and two rounds of interviews with a total of 14 respondents from relevant departments at Scania CV and Traton to source empirical data on the current visions of an aftersales system for driverless trucks. The first round of interviews was mostly with upper managers and the second round with lower managers.

According to the findings, there are various ambiguities and challenges regarding both technological and business-related aspects of the aftersales system design for driverless truck applications, establishing a need for collaboration between the concerned departments. There are differing visions between departments and within them. Various theories can support the collaboration process and handle ambiguity. Based on the empirical findings, a dissection of the current status quo is presented, and a discussion of how more favorable conditions for aftersales development can be induced.


**Keywords:** Driverless trucks, Aftersales, Product-service system, Ambidextrous organizations, Change management, Visions, Ambiguity.

# FOREWORD

We would like to extend our heartfelt thanks to everyone who made this work possible. First and foremost, a huge thank you to our supervisor, Lina Rylander. Your fantastic support from start to finish of this thesis project has been invaluable, and we would not have achieved the results that we have without your guidance. This semester has been full of interesting learnings and enjoyment, and we feel privileged to have gotten the opportunity to conduct our thesis at Scania CV and to support your research.

We also want to show our appreciation to Jonas Legnell and Håkan Warnquist from Scania CV for reviewing the report. Your insightful feedback and warm welcome at the beginning of the project really set us on the right path.

A special thanks goes out to the respondents from the workshops and various departments within Scania CV and Traton. Your strong commitment during the visits and interviews was crucial in providing the data and insights we needed. We truly appreciate your time and effort.

We also want to express our gratitude to our family and friends for their moral support and valuable discussions, which sparked new ideas and approaches. Your encouragement kept us motivated throughout our thesis project.

In the end, this thesis is the result of the collective effort and support from many individuals, and we are deeply grateful to each and every one of you for your contributions.

*Douglas Danfors*
*& Gustav Olson*

Södertälje, June 2024

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# 1 INTRODUCTION

*This chapter presents the background of the thesis project, along with its purpose, research questions, and delimitations.*

## 1.1 Background

Autonomous vehicles, or driverless vehicles, are defined as motor vehicles that can operate without a driver on a sustained basis with the help of a driving automation system (SAE International, 2021a). The vehicle industry has had driverless vehicles in its scope for a long time, and today, immense effort is devoted to making the technology a reality. Driverless vehicles have also gained much attention in the research community, establishing a field where much effort is dedicated to understanding how a shift to driverless technology will affect the automotive industry and society (Bathla et al., 2022; Hudson et al., 2019).

To grade and evaluate driverless vehicles based on their automated driving capabilities, SAE has established a framework with six levels of different autonomy requirements (SAE International, 2021b). According to the SAE framework, fully autonomous vehicles are represented by SAE levels 4 and 5, as the automated driving features do not need support from a physical driver at any point to operate the vehicle. See Figure 1 for an illustration of the different autonomy levels and what requirements each level contains.

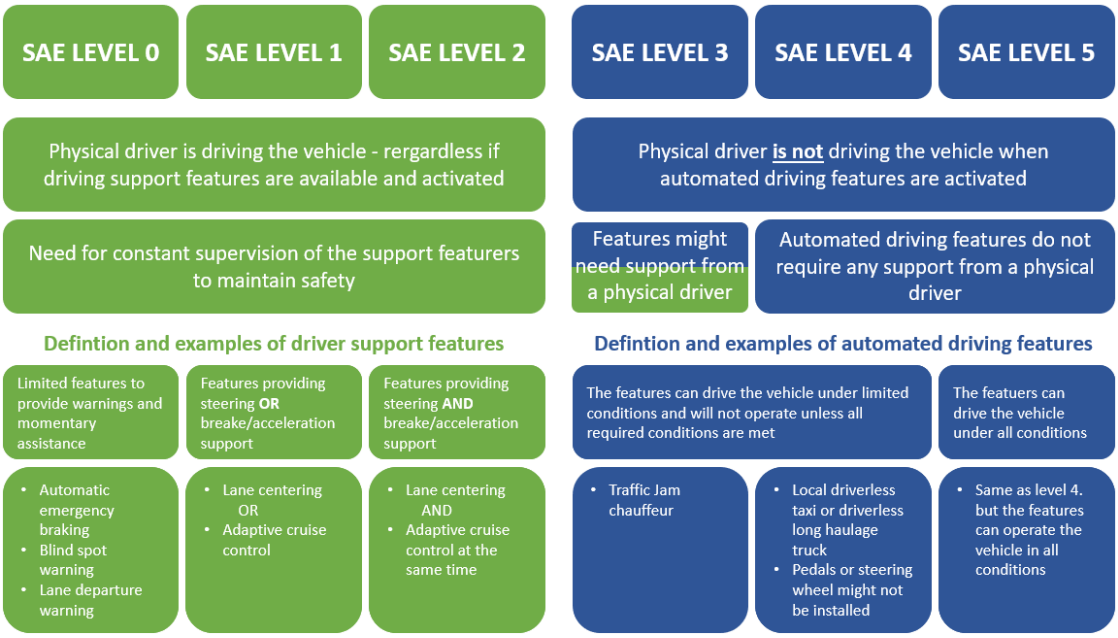



*Figure 1: Illustration of the six SAE levels for driverless vehicles. Inspired by* (SAE International, 2021b)

Today, there is an ongoing change in the truck industry toward the development of driverless trucks, and multiple incentives, such as “more hours of service” and “better fuel efficiency”, are driving the development forward since human factors limit the utilization rate a truck can have and how the vehicles are operated (Kristoffersson & Pernestål Brenden, 2018). With a transition toward driverless trucks, the driver's role will inevitably remain in the form we are familiar with today, thereby undergoing a drastic change. The revision of truck operation can potentially cause social problems as the driver is removed and labor opportunities vanish (Kristoffersson & Pernestål Brenden, 2018).

However, others state that the impact on the labor market for drivers will be minor as the transition from trucks to autonomous ones will be slow (Andersson & Ivehammar, 2019), ultimately establishing uncertainty regarding the impact of driverless trucks.

To concretize the statements above, Scania CV sees its autonomous transport solutions as contributors to a safer, more efficient, and sustainable future (Scania Group, n.d.a). They see an increasing demand for autonomous transport solutions, and significant steps are being taken in both technology and legislation. The company focuses on hub-to-hub (long-haulage transports between logistics terminals) and mining transports, and its parent company, Traton, has estimated that autonomous driving will stand for 78 billion km of road transport in 2026 (Scania Group, n.d.a). Furthermore, it is stated that driverless trucks will be more expensive initially and could approximately increase cost by one and a half to two times compared to today's vehicles. However, by removing the physical driver, extensive cost savings can be made over time (Scania Group, n.d.a). Other sources agree with this statement, expressing that the benefits and opportunities with driverless trucks are that they eliminate driver-related costs (Andersson & Ivehammar, 2019; Kristoffersson & Pernestål Brenden, 2018).

Research has been conducted regarding driverless trucks and business models, where it is presented that truck manufacturers will offer their customers a servitized offering instead of solely providing the truck as a product (Monios & Bergqvist, 2020). The incorporation of servitized offerings is a common phenomenon within a wide array of different technological industries and allows for higher customer integration, value proposal, and potential for revenue. This is often referred to as Product Service Systems (PSS), where products and services are integrated into the same offering to create more customer value (Sundin et al., 2009). A key aspect of PSS design is having a wider focus than pure product performance and taking the complete life cycle of the product into consideration, consisting of manufacturing, usage, maintenance, and end-of-life activities, where improvements in maintenance activity design especially can stand for great revenue increase (Sundin et al., 2009).

Therefore, it can be argued that fault handling and maintenance will play a bigger role in driverless truck offerings as it will be a central part of the value proposal and value capture. However, Tao et al. (2022) present that with the rapid development of autonomous driving, there are rising challenges regarding short-term maintenance planning for driverless trucks. Rylander (2023) highlights that there will be challenges with replacing truck drivers as they have a crucial role in the fault-handling process of today's system. Rylander (2023) also states that there is a limited amount of research conducted regarding how driverless trucks will affect uptime, which is troublesome since uptime is a vital factor for vehicle operation.

Uptime itself can be defined as the time that a vehicle is in upstate and is ready to be used (Bernspång & Kali, 2011). The opposite of uptime is downtime, which can be defined as the time when the vehicle is unavailable for usage due to maintenance activities (Bernspång & Kali, 2011). Together, uptime and downtime represent the planned operating time which is the total time that the vehicle is intended to be used (Bernspång & Kali, 2011), and uptime is usually promoted by sufficient fault management and maintenance activities (Rylander, 2023). In the case of fault handling for driverless trucks, it is important to consider the truck driver's role in the case of malfunctions, as essential detection and fault management capabilities will vanish as the driver is removed (Rylander & Englund, 2024). Therefore, it will be important to understand how the needs of fault management processes will change with the introduction of driverless trucks to promote sufficient levels of uptime (Rylander & Englund, 2024).

The current lack of insight regarding maintenance aspects for driverless trucks shows that there is a need for further research to gather knowledge of how the transition should be dealt with and how aftersales activities and fault handling should be adapted to promote uptime for driverless trucks. In this thesis, the aftersales system is defined as a product-service system where the capabilities and activities that enable driverless truck uptime are core elements. This includes the

fault handling process (diagnosis and repair), maintenance, and the aftersales offering. The aftersales offering covers the business aspects of aftersales capabilities and activities and is primarily regarded as the value offered to the customer.

Visions are crucial for technology development to cater to development activities with uncertain and radical characters, as without them and without people following them, technology would not be developed (Bemler, 2023). Bemler (2023) also explains that vision regarding driverless trucks is an interesting area to conduct research within as the visions are created by different intersecting areas: expectations, organizational future visions, and beliefs about technology's role and impact on society. In this thesis project, visions are defined as desirable states of the future that set the organization's direction. Since limited focus has been directed toward aftersales development for driverless trucks, establishing a clear and well-defined vision seems to be essential to guarantee a sufficient result of the development strides.

## 1.2 Purpose

This thesis project aims to investigate expected challenges and current visions of the aftersales system for driverless trucks in a hub-to-hub context, focusing on Scania CV and Traton. The study intends to identify potential obstacles in developing aftersales activities for driverless trucks and proposes guidelines for managing the development. The guidelines are presented by implementing Lewin's three-step model, a foundational change management framework for organizational change. This thesis project also aims to support academia within the driverless truck field by combining relevant theoretical frameworks with empirical data from one of the more prominent actors within the truck industry to target the current and future needs of aftersales-related aspects for driverless trucks.

To fulfill the purpose presented above, the following research questions will be examined:

> RQ1: *What are the current visions of the aftersales system for driverless trucks within the truck industry/Scania CV, and what challenges can be foreseen in driving the development of the system?*
>
> RQ2: *How can guidelines be formulated to address the challenges identified in developing the aftersales systems for driverless trucks, with the help of change management theory, specifically utilizing Lewin's three-step model?*

The deliverables of this thesis project are to present a state-of-the-art description of the existing challenges and vision for the development of an aftermarket system for driverless trucks and propose guidelines for how these challenges can be addressed.

## 1.3 Delimitations

These are the following delimitations of the project:

- This research study was a thesis project for the Master of Science program in Integrated Product Design, with the track Innovation Management and Product Development, at KTH.
- The duration of the project was approximately 20 weeks.
- The driverless truck aftersales system is regarded as a product-service system.
- The scope was narrowed to investigate the driverless truck aftersales offering in a hub-to-hub context.
- In this thesis, a vision is a shared representation of the desired future and can cover both short- and long-term perspectives.

- The thesis project is an exploratory research study, and the findings should be considered as guidelines or proposals for aftersales development for driverless trucks.
- This thesis will primarily cover automated trucks of SAE levels four and five (SAE International, 2021b). With these levels, there are no physical drivers present in the vehicles.
- Since Traton is a large organization esatbilshed in a wide variety of geographic locations, the scope of this thesis project is on a global scale. However, as the focus company of the study primarily is Scania CV, most of the gathered insight are sourced from the outlook of the Swedish and European markets for driverless trucks.

# 2 FRAME OF REFERENCE

*This chapter summarizes relevant literature and research fields for the thesis project. The citations and literature presented in this chapter will be used to analyze and discuss the findings from the empirical data.*

## 2.1 Smart products & services and product-service systems

This section defines smart products and services and discusses how organizations might face challenges when transitioning their product portfolio to incorporate new software-heavy solutions. It also describes product-service systems as a business model innovation phenomenon and how organizations can manage the merging of products and services as a complete offering. Finally, a connection between product-service systems and driverless truck literature is established to illustrate how the different research fields are related to each other.

### 2.1.1 Smart products and smart services

Smart and connected products are one of the biggest trends in the technology industry today, and many manufacturing firms are trying to complement traditional hardware products with software capabilities (Porter & Heppelmann, 2021). According to Porter & Heppelmann, smart and connected products consist of three different component categories: physical components (mechanical or electrical parts), smart components (sensors, embedded systems, and digital user interface), and connectivity components (ports and antennas). Smart products are also specialized in specific types of data processing, providing insightful data to their users (Lee & Abuali, 2011).

By combining software and hardware in this manner, new product capabilities are achieved, such as real-time monitoring, remote accessibility, and ware-profile data, which has changed a wide array of industries radically, ranging from home electronics to industrial manufacturing equipment and automotive solutions (Porter & Heppelmann, 2021). Smart products have the potential to understand the customer, as data is sourced and analyzed to provide valuable insights into the customer's product utilization, establishing a need to involve the customer in the development of both industrial and software design attributes of the product (Lee & Abuali, 2011). As a result, smart and connected products require manufacturers to revise their customer interaction as there is an opportunity to integrate with the customer’s operation due to the possibility of sourcing, analyzing, and providing insightful data to the customer (Porter & Heppelmann, 2021). Furthermore, smart and connected products have a holistic effect on the complete value chain, changing the function of product development, IT, manufacturing, logistics, marketing, sales, and aftersales due to the added product capabilities, establishing a need for cross-functional collaboration to enable a sufficient and streamlined adaption of the organization’s departments (Porter & Heppelmann, 2021), while also enabling great opportunities for revenue increase by integrating service into the product offering (Lee & Abuali, 2011).

Accordingly, the aftersales offering for smart and connected products is a specific field where drastic revisions can be made to increase profit (Porter & Heppelmann, 2021). According to the authors, service and maintenance for long-lived products can generate substantial revenue but also be improved as conventional service activities are often inefficient in terms of profit ratio. With smart and connected products, there is room to innovate service activities and aftersales-related business models since the product technology enables preventive and remote maintenance (Porter & Heppelmann, 2021). Allmendinger & Lombreglia (2005) add that “smart services” completely revise the aftersales offering as maintenance shifts from reactive to preventive. By utilizing the sensing and computational capabilities of smart products, maintenance programs can

be based on real-time monitoring of immense amounts of data, making sure that maintenance for preventing malfunctions only is conducted when needed, meaning smart services can generate a new level of customer value and becomes a market differentiator (Allmendinger & Lombreglia, 2005; Lee & Abuali, 2011).

Porter & Heppelman (2021) imply that the preventive service attributes can be seen as "a new era of lean," enabling processes throughout the whole value chain to be more streamlined as waste can be eliminated to a further extent while profit is optimized. However, this also creates a need for new expertise as the focus is shifted from mechanical engineering and selling products and maintenance to software engineering and offering a service where the amount of guaranteed uptime is the most important factor (Porter & Heppelmann, 2021).

### 2.1.2 Product-service system

Staying relevant in changing industry environments can be a great challenge, and to do so, firms tend to rely on business model innovation to maintain their competitive edge (Velamuri et al., 2013). In the context of technological change and the introduction of smart products, a common action is to revise product portfolio structures to allow for servitization and integrate services and products in the same offering, creating a product-service system (PSS) to widen the value proposal scope (Velamuri et al., 2013).

The PSS concept has evolved immensely since its first introduction in 1999, primarily focusing on enabling efficient data transmission for mobile internet transmission toward incorporating smart PSS offerings where software is embedded in the base product to allow for servitization and value creation (Zheng et al., 2019). Two major contributors to the development of the PSS concept were the upbringing of service-dominant logic, focusing on providing knowledge and skill as an offering itself to build relationships (Vargo & Lusch, 2004), as well as the establishment of Tukker's Eight Types of PSS (Zheng et al., 2019). According to Tukker (2004), PSS can be divided into three main categories, focusing on either product-oriented services, use-oriented services, or result-oriented services. These three categories include various business model proposals for PSS offerings, functioning as a framework for applicable business strategies for organizations trying to pursue servitization (Tukker, 2004; Zheng et al., 2019).

The incorporation of PSSs is usually, and ideally, focused on increasing customer satisfaction, making the offering customer-oriented rather than technology-driven (Velamuri et al., 2013). For manufacturing firms, servitization allows for higher customer integration but also an opportunity for horizontal exploitation in the value chain (Wise & Baumgartner, 1999), as well as revolving around the complete life-cycle of a product to establish new possibilities for revenue stream (Allmendinger & Lombreglia, 2005). One way of emerging towards a PSS-oriented business model is to introduce "Comprehensive Services", meaning traditional repair and maintenance offerings are integrated with the base product to, for instance, allow for a specific amount of guaranteed uptime (Velamuri et al., 2013). This changes the character of the offering as the conventional buyer-seller relationship transitions into a more partnership-like dynamic (Velamuri et al., 2013).

Mittag et al. (2018) further expand on the importance of having a balanced ratio between industry push and market pull when transitioning toward a PSS offering. To make the servitization of a product successful, firms must evaluate their product's technical capabilities (sensors, data processing, connectivity, and user interfaces) to see how service attributes can complement it and make it more attractive while also understanding what service concepts the market will demand and consider as valuable (Mittag et al., 2018).

Many manufacturing firms have, and are, revising their offerings to enable a switch from selling products to providing service solutions (Parida et al., 2014). However, there are some common challenges that transitioning organizations might face, making the revision of their offering

troublesome (Martinez et al., 2010; Parida et al., 2014). First and foremost, it is crucial to understand the customer needs to make the corresponding business model sufficient, targeting the customer demand(s) (Parida et al., 2014). Therefore, business model design should be conducted with tight customer collaboration, communicating the value proposition clearly to understand if revisions must be made to capture customer demand (Parida et al., 2014).

Furthermore, there is a need for an adaption of the organization's internal processes, capabilities, strategy, and culture to allow for a complete revision of a PSS offering (Martinez et al., 2010). Necessary measures could be to re-engineer established processes within the organization that do not support a service-oriented offering or implement cross-functional collaboration to make departments develop as a unit to support a servitized business model better (Martinez et al., 2010). Lastly, there is a need for improved supplier collaboration to allow for more service-oriented activities, as the organization might need to involve suppliers with additional expertise or resources to be able to deliver their servitized offerings (Martinez et al., 2010; Parida et al., 2014).

### 2.1.3 Driverless trucks as a product-service system

Utilizing a PSS structure could be relevant for OEMs in the road freight industry as the transition toward driverless trucks might lead to an immense shift in business model architecture and customer needs, as driverless trucks might not be sold or operated in the same manner as the conventional ones of today (Monios & Bergqvist, 2020). Monios & Bergvist (2020) argue for the rise of new business cases where OEMs may own their own trucks and offer transportation of goods instead, which changes the business model radically, approaching a transport solution where products (the truck) and services are offered as a package instead of being separate affairs (PSS). As the software capabilities of the vehicles will play an even greater role in driverless road transportation due to the need for remote access and monitoring, novel ways of utilizing the embedded systems and offering a well-rounded system solution will be essential to gaining market share (Monios & Bergqvist, 2020).

However, the future of driverless trucks is uncertain to a wide extent, as ambiguity regarding the effects of its establishment is yet to be discovered (Engholm, 2021). According to Engholm, the primary driver for driverless trucks is a reduction in the total cost of ownership (TCO) for freight operators, compared to conventional trucks, as driver-related costs are eliminated. This could lead to more cost-effective road freight and, thereby, a competitive advantage in hub-to-hub operations. Yet, the business model perspective will need to be further investigated as a thorough picture of the market acceptance and specific business case for driverless trucks is still to be defined (Engholm, 2021). Also, further research is required to investigate the sustainability impacts and societal effects of driverless trucks to fully understand how the transition will affect the transportation industry, as these factors might have influences on both technological and business model development (Engholm, 2021).

## *2.2 Ambidextrous organizations*

This section introduces ambidexterity as an organizational structure and how it can be implemented to manage radical innovation. It includes an introduction to the purpose of ambidextrous organizations, followed by how the organizational structure benefits radical innovation capabilities and what challenges an organization might face by adopting the structure.

### 2.2.1 Purpose of ambidextrous organizations

As implied previously, OEMs might have to revise their current product portfolio and business models to benefit from the value-adding possibilities of driverless trucks. In order for organizations to remain competitive, there is a need to allocate resources toward both incremental and radical innovation areas, developing completely new products or services while also

improving their current portfolio (Birkinshaw & Gibson, 2004). An ambidextrous organization devotes effort to both exploitive and exploratory tasks to maintain its competitive edge but also creates breakthrough ideas by segregating exploratory departments from the traditional units, enabling freedom and creativity for teams working with novel technology and radical innovation (Birkinshaw & Gibson, 2004; O'Reilly III & Tushman, 2004). According to O'Reilly III and Tushman (2004), firms with ambidextrous structures have a nine times higher likelihood of developing novel and successful products and processes while also maintaining or improving their core business area, compared to firms that implement alternative organizational structures

Organizations operating in technology industries are affected by technological cycles, shifting between periods of incremental change and periods of rapid innovation (Abernathy & Utterback, 1988; Sanderson & Uzumeri, 1995; Tushman & Anderson, 1986). One of the most challenging aspects of organizational process management, and crucial capabilities to stay competitive in a changing business environment, is having the ability to sense these cycles and thereby understand when and how to exploit their core business to maximize short-term success, and when to explore new business areas to secure future market share (Benner & Tushman, 2003). In ambidextrous organizations, process management is isolated in subunits, enabling departments to operate independently of each other to let them develop their own processes, cultures, and structures (Benner & Tushman, 2003; O'Reilly III & Tushman, 2004). This reduces the risk of having negative disturbances between units focusing on incremental and radical development, making sure efficient and effective teams do not hinder the creativity and innovative capabilities of others or vice versa (O'Reilly III & Tushman, 2004). The only commonality between the different units is that they are integrated at the top management level, as their budgets and resources have the same origin (O'Reilly III & Tushman, 2004).

### 2.2.2 Ambidextrous organizations and radical innovation

The greatest driving force for having a designated department focusing on novel technology and radical innovations, referred to as an autonomous team, is that the uncertain and complex attributes of radical innovations require completely different characteristics compared to the capabilities that are beneficial for incremental innovation (Patanakul et al., 2012). An autonomous team can be compared to a start-up within the organization, having its own processes and structures (O'Reilly III & Tushman, 2004). To handle radical innovation successfully, the team must have a culture that supports exploratory learning rather than effective task deployment, reward systems and performance metrics that are tailored for an uncertain and development environment, as well as personnel with appropriate skills for the task at hand, which can vary immensely from the optimal conditions for conducting incremental innovation (Patanakul et al., 2012).

The addition of agile work methods, originally introduced for software development, is also a common characteristic for autonomous teams, as it involves iterative processes, customer integration, and high adaptability for change, fulfilling the demands of uncertain development well (Highsmith & Cockburn, 2001; Mikalsen et al., 2019). Customer or partner integration can be achieved by implementing open innovation and inter-organizational collaboration, serving as a useful tool for radical innovation as new knowledge from adjacent market fields can be utilized in development activities (Almirall & Casadesus-Masanell, 2010).

However, the implementation of agile and autonomous teams in big organizations where cross-functional collaboration is needed can imply issues if the autonomous team has too many external dependencies as it makes work slow (Mikalsen et al., 2019). Also, if inter-organizational collaboration is utilized, the external partner network must be constructed with diligence, as proximities between the partnering organizations will have major effects on the quality of the collaboration output (Knoben & Oerlemans, 2006). For instance, inter-organizational

collaboration for increased innovation capabilities must include equal levels of geographical, organizational, and technological proximity in order for the firms to understand each other and establish sufficient conditions for fruitful collaboration (Knoben & Oerlemans, 2006).

Furthermore, the drawback of autonomous teams is their independence from the remaining organization, as management might have limited control over the unit due to its high grade of flexibility (Patanakul et al., 2012). Birkinshaw & Gibson (2004) also emphasize that structural ambidexterity, where subgroups are isolated from each other, can be insufficient if the evaluation of R&D development output is conducted by top managers with no direct connection to the specific subgroup, as misalignments in focus can lead to misleading decisions. Structural ambidexterity can also be rather expensive since it might be difficult to share the resources and capabilities of the autonomous department with other units or functional groups (Patanakul et al., 2012). Also, complete separation of autonomous teams from their parent organizations might inhibit successful innovation output, as some connection or support from other departments and core functions is essential to manage radical innovation sufficiently (Patanakul et al., 2012). Therefore, organizations must balance the connectivity and independence of autonomous teams to guarantee enough flexibility to not inhibit necessary capabilities for radical innovation, while at the same time also enabling resource and knowledge exchange with other departments within the organization if needed (Patanakul et al., 2012).

Another important aspect of the success rate for ambidextrous organizations is a clear and compelling vision communication among the senior or top-level managers, showcasing the need for ambidexterity throughout the whole organization (O'Reilly III & Tushman, 2004). If done sufficiently, the vision will be spread across all subunits, establishing an understating of why the coexistence of exploration and exploitation is important for the organization's current and future success.

## *2.3 Visions within organizations*

This section will cover how a "vision" can be defined, how visions are utilized within organizations, the common challenges associated with visions, and technological paradigms and trajectories.

### 2.3.1 The definition of a vision

Organizations use visions to create a shared understanding of the future, setting a direction for what lies ahead to motivate the employees to reach that shared understanding (Kaiser et al., 2021). There are various definitions of the word "vision", but one presented by Bemler (2023, s. 15), who researches the future of driverless trucks, is the following: "Visions: Represent the stories told about the future that contain goals and aims for the future of self-driving vehicles and the society in which they should operate.". However, in this thesis project, visions are defined as desirable states of the future that set the organization's direction.

Kotter´s eight-step model (1995) is well renowned and used by organizations to manage change. In his model, visions are a crucial part. According to Kotter, Visions can be described as a forward-looking image of what the organization wants to be in the future, and it should motivate and compel all involved in the change to steer them in the right direction. Without a clear vision, the organization risks going in the wrong direction (Kotter, 1995). Visions should include seven attributes (brevity, clarity, abstractness, challenge, future orientation, stability, and desirability) as these are connected to organizational performance (Haque et al., 2016). Furthermore, the vision must be inspirational and widely shared to prepare the organization for change.

### 2.3.2 Challenges regarding visions within organizations

There are challenges regarding creating and implementing visions, and one of the challenges is to combine the more visionary nature with practical and measurable goals within the organization (Mintzberg, 1987). If not, it can lead to skepticism and resistance toward the vision among the employees (Mintzberg, 1987). Another challenge when implementing visions in an organization is the organization's capability to handle the complexities of changing organizational structures, cultures, and behaviors, which greatly impact the implementation of strategic vision (Gratton, 1996). A research study was conducted with 60 middle managers from a large international organization regarding vision implementation challenges (Hodgkinson, 2002). The results indicated that barriers to a shared strategic vision were cultural and hierarchal issues as well as personal agendas of upper management. The middle managers emphasized a need for an organizational culture of learning, innovation, and collaboration to enhance the ability to create a shared strategic vision.

Within organizations, the departments can create and implement individual visions due to having specific goals that are important for that department. Lewis and Clark (2020) recognized this dilemma as, in large organizations with sub-units, sub-organizational visions exist that can differ from the organizational vision. The differences are affected by the company's structure; the sub-units that are farther from the organization's core are inclined to create visions that have no direct connection to the organizational vision. However, organizations can have mechanisms that ensure that the different visions complement each other like a puzzle (Lewis & Clark, 2020).

### 2.3.3 Technological paradigms and technological trajectories

Organizations follow different directions when they work with technological development, which can be described as technological trajectories. The technological trajectories are affected by the existing technological paradigms, which are defined as the beliefs and knowledge regarding the problem and solution that exists within the organization (Dosi, 1982). Discontinuous innovations affect this, as these form new technological paradigms. New technological paradigms occur due to a combination of different matters, new technologies appearing, financial factors, and obstacles on existing technology paths (Dosi, 1982). To handle technological paradigm shifts, organizations should focus on “co-evaluation” between participants who are affected by the shift and conduct experiments that may have a high failure rate to learn more about the paradigm shift (Bessant et al., 2005). The authors also argue that organizations should collaborate with researchers to gather insights and learn more regarding the new technological paradigm.

## *2.4 How to manage ambiguity*

This section will introduce the concept of ambiguity and how organizations can use the sensemaking process to handle ambiguity.

### 2.4.1 The definition of ambiguity

Ambiguity is when there are several different interpretations of a matter existing, which confuses (Weick, 1995). Weick further explains that this critically affects rational decision-making and that the problem in situations where there is ambiguity is not that we need more information to understand. Instead, the problem is that with even more information, the misunderstandings might not be cleared up. Another common concept is uncertainty, which is quite similar to ambiguity. However, there is a distinct difference between uncertainty and ambiguity (Weick, 1995). When there is uncertainty, the interpretations are too few, and more information is needed, and when there is ambiguity, there are too many interpretations, and instead, the right kind of information is missing (Weick, 1995). There is a great challenge in finding what information is needed, as the

ones that are affected by ambiguity do not know where to start and what data are relevant; they are unsure about what questions to ask to tackle the ambiguity (Weick, 1995).

### 2.4.2 The sensemaking process

Sensemaking is an approach that can be used for many things. However, it is also an appropriate way of processing ambiguity (Weick, 1995). Sensemaking can include comprehending different matters, forming meaning, and making a common understanding (Weick, 1995). Retrospection is a crucial part of sensemaking, as people form their understanding with the help of previous experiences and knowledge (Weick, 1995). Weick has divided sensemaking into seven properties, which can be summarized in the following manner (Helms Mills et al., 2010):

- **Grounded in identity construction:** The affected individuals and groups must figure out their identities in the upcoming environment and how they will be affected.
- **Retrospective:** The individuals and groups look back at previous experiences to retrieve data that supports the clarification of the new environment.
- **Enactive of sensible environments:** Creating an organized setting with the support of sensemaking, the aim is to turn a confusing situation into a clearer and more structured one.
- **Social:** The individuals and groups form their interpretations of the new environment through social interactions.
- **Ongoing:** The individuals and groups must constantly work with sensemaking as sense only is temporary and is affected by disruptions.
- **Focused on and by extracted cues:** The individuals and groups should seek cues; they are the material that enables interpretations of the environment
- **Driven by plausibility rather than accuracy:** Focus on what seems most reasonable as sense only supports the individuals and groups to get enough clarity of the environment and not so that they fully understand it.

These properties are not set in stone, and some can be skipped as the process of sensemaking is affected by feedback loops, and some of the properties can be managed simultaneously (Weick, 1995). However, sensemaking is pretty ambiguous as it can be difficult to grasp. Sandberg Tsoukas (2015) have described sensemaking in an organizational context with four major components to clarify. Firstly, sensemaking is activated by ambiguity. It can be ambiguity connected to both lesser and greater events that are either planned or unplanned. Secondly, sensemaking occurs through specific processes: creation, interpretation, and enacting of ideas. Thirdly, this results in specific outcomes due to the practical implementation of ideas. Fourthly, sensemaking is affected by situational factors, such as language, identity, and technology.

As presented, ambiguity acts as a trigger for sensemaking (Sandberg & Tsoukas, 2015). Therefore, it is an appropriate process to solve the ambiguity. Sensemaking can support organizations that deal with ambiguity as it helps the affected individuals to look for a “why”, accept what seems reasonable, and move forward (Weick et al., 2005). As mentioned, ambiguity confuses, and it is difficult for those involved to know what data that is relevant. In these situations, the sensemaking process should utilize face-to-face interaction with multiple cues to reduce various interpretations of the information (Weick, 1995). The author proposes that meetings and direct contact containing many different triggers are more appropriate than formal information systems and reports to reduce ambiguity and that doing the opposite is a common mistake. Weick et al. (2005) have presented that sensemaking provides several points of understanding for organizations:

- The analysis of how individuals make sense shows that small changes can lead to great changes over time.
- Actions are usually made before the understanding of the environment has been made.
- Exploring what individuals do before making a decision.
- Creates an understanding of how individuals can change the environment through actions.
- Gives the organization a chance to include and highlight meaning and thoughts in how they work.
- There is often a separation between thinking and actions, which sensemaking erases.
- Having the perspective of viewing mistakes as learnings. Making reasonable guesses, taking small steps, and adapting as you go are good enough to reach your goal.

To enable the process of sensemaking, the organization should aim for their employees to learn to stay resilient in different situations, see limits as set by themselves, focus on what seems reasonable, be action-oriented, use retrospective to understand new environments, and form ideas that motivate others (Weick et al., 2005). The authors argue that these are small actions for the individual. However, they can have great impacts. It is also important for the organization to consider the politics of sensemaking. There is a risk that the lower-level employees don't get the mandate to shape the sensemaking within the organization. Instead, it risks being dominated by upper management, which can affect the results of the sensemaking process (Maitlis & Sonenshein, 2010).

## *2.5 Change management*

This section will introduce change management and present the well-renowned "Lewin´s three-step model". It will also present the importance of transformative leadership and how to perceive resistance to change.

### 2.5.1 Lewin's three-step model

In uncertain and ambiguous environments, there might be a need for organizational change. Change is affecting all organizations, and therefore, there is a need for organizations to be able to manage it (Todnem, 2005). The author argues that this is needed to stay competitive in business environments that are constantly changing. There is a lot of literature regarding change management, and many different frameworks have been presented and utilized since the concept's origin. However, there is a need for more research in the field of change management since many studies lack empirical data, and many of them contradict each other (Todnem, 2005). One well-known framework for change management is Lewin's three-step model, and Lewin´s ideas have acted as a foundation for the development of modern change management practices (Burnes, 2004).

Lewin´s three-step model has been widely used to manage organizational change (Burnes, 2004). However, Lewin did not plan for the model to be viewed independently from the rest of his research, which included field theory, group dynamics, and action research (Burnes, 2004). The author means that the three-step model is supposed to be used together with these research areas to grasp better how change for groups, organizations, and societies takes place and can be managed. The model is originally presented by (Lewin, 1947) as a three-step procedure. However, it has been examined and presented more comprehensively by other researchers over the years. These are the following steps in Lewin´s three-step model:

1. Unfreeze: Identifying and destabilizing quasi-stationary equilibrium one (Burnes, 2020). Prepare the organization to accept a need for change, spread awareness of the need, and present the cons with the current state (Schein, 1996).

2. Change: Implement new processes that affect the behaviors of those involved (Schein, 1996). The author argues that the changes need to be clearly communicated, which puts high demands on leadership.
3. Refreeze: Reinforce quasi-stationary equilibrium two (Burnes, 2020). Ensure that the changes have been implemented well and support them constantly so that they become part of the organization's foundation (Schein, 1996).

The movement between the stages of the process is steered and controlled by group dynamics, democratic participation, and choice (Burnes, 2020). The author also means that all steps are iterative, regarding the following actions, finding facts, performing an action, and searching for more facts, see Figure 2.

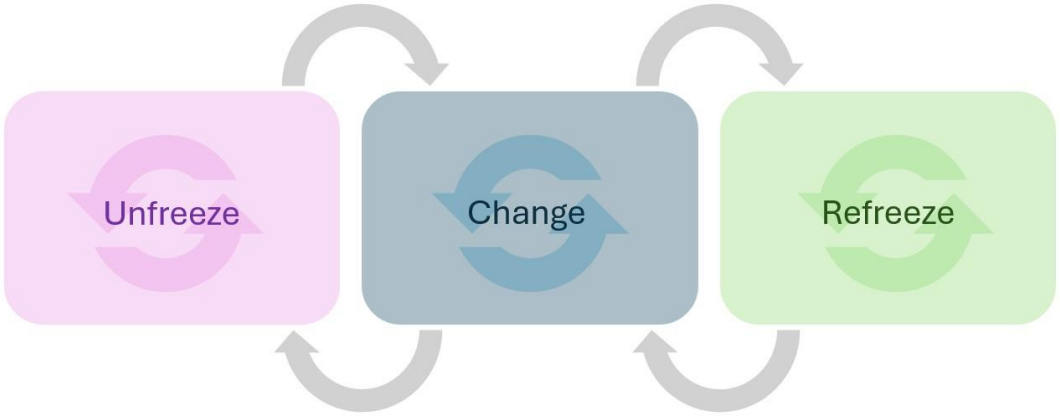



*Figure 2: Illustration of Lewin´s three-step model, inspired by* Burnes (2020).

Lewin´s three-step model has faced a lot of critique during the years of its existence. Common critiques against it have been the following: it is too simplistic to handle the continuous change of organizations; the model is only appropriate to handle incremental changes; it does not consider how power and politics affect organizations; and it highlights that a top-down management approach is more appropriate than a bottom-up management approach when dealing with change management (Dawson, 2019). However, Burnes (2004) argues that Lewin's perspective on change management was that organizational change is a complex process and that there is a need for an iterative learning process when undergoing it. He identified that you need to consider what power structures exist within the organization and that the motivations behind change can come from all directions, not only from the top of the organization. The author means that most of the critiques have been misinterpretations of Lewin´s work.

### 2.5.2 Transformative leadership

In Lewin´s three-step model, leadership has a central role throughout the steps, and the transformational leadership style has been proven to be appropriate (Hussain et al., 2018). In this leadership style, the leader collaborates closely with employees, shares knowledge, and involves them in making decisions at the organizational level (Hussain et al., 2018). The authors present that employee involvement is the most important aspect to advance through the steps of Lewin´s three-step model and highlight that knowledge sharing is an important tool to use in the unfreezing step, as you need all of the organization on board that there is a need for change.

To encourage innovative behavior amongst employees, it is important to improve the individual perception of the workplace as being encouraging regarding taking initiative, which transformative leadership effectively can enable (Michaelis et al., 2010). The authors also argue that supportive factors that encourage a climate that enables initiative amongst the employees should be included in the organization's reward system. They further state that companies should provide education on transformative leadership to their management personnel, which can improve their capabilities regarding the commitment of their colleagues, which results in innovation implementation behavior (Michaelis et al., 2010). The benefits of transformative leadership are also presented by Nasir et al. (2022), as they state that it motivates employees to be more creative, which affects the performance of the organization.

### 2.5.3 Resistance to change

Resistance to change is important to have in mind when conducting change management, as it is a part of every change process (Pardo Del Val & Martínez Fuentes, 2003). The authors argue that resistance to change can be seen as a phenomenon that acts as a barrier to change, which is to maintain the situation of today. However, this view of resistance to change is not shared by Cinite and Duxbury (2018). Instead, from their perspective, it is primarily about an employee who raises concerns about changes that affect them. The change attitude can vary between being very supportive and being strongly resistant. However, there are difficulties in identifying the attitudes within an organization, as the resistance toward change can be passive (Cinite & Duxbury, 2018). The two authors have created a proposal for measurements of resistance to change that organizations could utilize. The first one is “Commitment to organizational change”, which shows that employees who work in a supportive environment and have low job stress levels are more committed to organizational change and can engage in behaviors like encouraging and promoting others regarding the change. The second one, “Voicing concerns about change”, shows that resistance is not solely negative, as some employees who raise their voices feel an obligation towards the organizations that they feel supported by to seek clarification of the change and propose suggestions for improvements. Cinite and Duxbury (2018) argue that it is the organization's responsibility to ensure the employees are committed to organizational change and that they are encouraged to voice their concerns about change to create a supportive environment for the employees.

# 3 METHOD

*This chapter presents the research methodology of the thesis project, including the different methods used during its duration.*

## 3.1 Research methodology

The purpose of this thesis has been to investigate the visions established within the transportation industry regarding the aftersales aspects of driverless trucks in hub-to-hub contexts. Since the field of study lacks both scientific rigor and practical applications, the thesis was conducted as a qualitative study in an explorative manner to promote the sufficient collection of empirical data. Qualitative studies are appropriate for this kind of context (Creswell, 2009).

This thesis can be categorized as a qualitative research study, as data collection was conducted through semi-structured interviews. The aim of this study was to investigate the different visions of key individuals within an organization. The qualitative approach lets the researchers gather data from the perspective of the participants or respondents, while a quantitative research approach is from the perspective of the researcher (Bryman & Bell, 2017). Also, quantitative research results in robust data that have one interpretation, compared to qualitative research, which collects rich data that can have many different interpretations (Bryman & Bell, 2017). However, qualitative research has been criticized for lacking the structure and stability needed when conducting research (Gioia et al., 2013). Therefore, the authors have developed an approach to follow when conducting inductive research that supports creativity and has a systematic rigor in qualitative, grounded theory research. The approach presented in the article consists of the following phases:

- Research design: Choose direction and formulate research questions with the help of literature.
- Data collection: For example, collect raw data through interviews. Be flexible regarding the design of the interview guide.
- Data analysis: Data coding constructs first-order terms (informant-centric) and groups them to create second-order themes (theory-centric). If appropriate, the second-order themes are sorted with overarching theoretical dimensions, creating a data structure.
- Grounded theory articulation: Find relationships within the data structure. Change the fixed data structure into a flexible and evolving theory model. Review more literature to better define and understand new ideas and their connections.

Another essential component in qualitative research is the implementation of iterative elements in the research process, which enable revisions of the research scope, literature review, empirical data collection, and data analysis (Polkinghorne, 2005). Therefore, to allow for continuous improvement along the thesis project, iteration of process stages was made if necessary to achieve a sufficient outcome. Polkinghorne (2005) also emphasizes that the quality, validity, and trustworthiness of qualitative research depends on the relevance of the data sources in relation to the research topic, in this case, how much knowledge the respondents acquire regarding the thesis topic. Therefore, sources should be chosen with care to guarantee a sufficient amount of expertise in order to allow for qualitative findings and analysis of both richness and depth (Polkinghorne, 2005).

Two steps that are important within qualitative research are coding and analyzis of the empirical data. Bryman & Bell (2017) express several guidelines for coding qualitative research. It is important to start with the coding early in the project and continue to work with it throughout the project. Go through the material multiple times, by firstly only reading through it, then secondly

generating codes. Review the codes generated in a critical manner and connect different codes that have a relation. Regarding analysis, (Bryman & Bell, 2017) presented recommendations for analyzing the material and finding themes. Some examples are that researchers should search for repetitive codes and topics, similarities and differences within the material, and material that is related to theory and existing research. These guidelines and recommendations regarding coding and analysis inspired the researchers of this thesis project.

To concretize and illustrate the research methodlology of this thesis, the double diamond model was utilized. The double diamond model is an iterative approach originally developed by the British Design Council (the Design Council, n.d.). The framework consists of divergent and convergent stages where emphasis first is directed toward exploration and to later narrowing the scope as there is enough insight to focus on specific elements of the problem (Tschimmel, 2012). The concept of the framework is valid for illustrating the iterative attributes of the research methodology, as redoing stages in qualitative research might be necessary to achieve a desirable result (Polkinghorne, 2005). The process consists of the following phases; see Figure 3 for an illustration of the research process:

- Firstly, a "discovery phase". It consisted of planning the project, a literature review, discussions with experts, field visits, and the first round of interviews.
- Secondly, a "define phase". The first round of interviews was transcribed, coded, and analyzed.
- Thirdly, a "development phase". The second round of interviews was planned and conducted. A more thorough literature study was also initiated during this phase to establish the frame of reference.
- Fourthly, a "decision stage". The second round of interviews was transcribed, coded, and analyzed. The material from the field visits and discussions with experts were also added to the analysis. The literature study was also completed. By the end of this phase, the analysis of all the materials was completed.
- Finally, the fifth and last phase consisted of a "delivery phase." Here, the focus was on compiling the final thesis report and preparing for the project presentations.

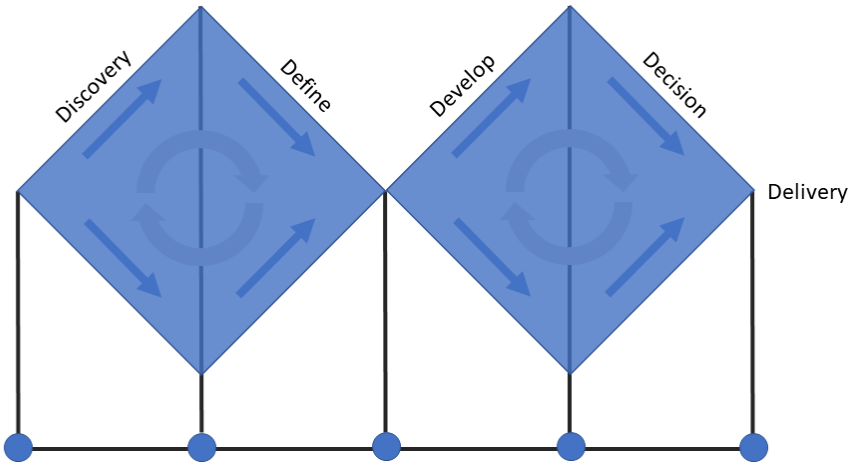



*Figure 3: The Double diamond model, inspired by* the Design Council (n.d.).

### 3.1.1 Assessment of methodology

This thesis project is a qualitative study, and there are critiques against these types of methods in a scholarly context. One critique is that qualitative research at times relies on minimal evidence and is too dependent on creative theorizing, which affects the validity of the findings as it influences the assertions made by the researchers (Gioia et al., 2013). The authors further explain that another challenge with qualitative research is to meet the strict requirements for theoretical progress in a scholarly context, as the methods are creative. This thesis project did not follow the grounded theory method meticulously. However, there are great similarities between the method followed and grounded theory. Therefore, critiques against the grounded theory can apply to this project as well. Bryman  Bell (2017) present that one con with grounded theory is that it can encourage the researcher to split up data and recover fragments while conducting the analysis, which can lead to the data being out of the wrong contexts. Grounded theory is a well-known method. However, it includes different approaches, and there is no clarity throughout the whole process regarding the structure, which confuses its users on how to use it in a correct manner (Bryman & Bell, 2017).

When conducting qualitative research, there are different criteria for assessing the quality of the research. Examples of criteria are credibility, validity, reliability, and justification of method (Bryman & Bell, 2017). Credibility, the researchers need to follow the rules that exist regarding the research and ensure that the involved individuals, for example, respondents, are aware of the result of the research so that they can approve the material they have supplied. Validity, as qualitative research often is conducted in a setting with a unique context, it is important that the research is adapted so that the results can be applicable in other settings. Reliability, it is important to in detail explain every step of the process for the research project so that the work can be properly evaluated. Justification of method, the researcher should not let their personal opinions and agendas affect the process or result of the research.

## *3.2 Research method*

In this section, the research method of the thesis project will be described.

### 3.2.1 Research setting

The research setting of this thesis project was at Scania CV and being a part of the research project System Transformation for Uptime for Driverless Vehicles (STIFF). Scania CV is a Swedish truck and bus manufacturer, as well as a brand and daughter company within the Traton Group. The headquarters is located in Södertälje, Sweden, and it has approximately 58,000 employees on a global scale (Scania Group, n.d.b). The STIFF project is a collaboration between academia and the industry to examine the requirements for Uptime for driverless vehicles. The discussions with experts, field visits, and interviews were all conducted within the Traton Group, with a main focus on the departments connected to aftersales and the development of driverless trucks.

### 3.2.2 Literature study

The literature study was used to gather knowledge regarding the research areas of the thesis project. It also supported the design of the interview guide. The databases used for the literature study were Web of Science, Scopus, and Google Scholar. A majority of the references that were used in the thesis project are research articles from different scientific journals; some of the references are also from books and websites. Specific keywords were used to search for references; some important keywords were Driverless trucks, Aftersales, Uptime, Product-service system, Ambidextrous organizations, Change management, Lewin’s three-step model, Visions, Technological paradigms, Technological trajectories, Ambiguity, and Sensemaking.  The

literature study was continuously updated throughout the project by adding and removing research areas and references, as the literature was adjusted based on the findings from the interviews.

### 3.2.3 Discussion with experts

The purpose of the discussion with experts was to further gather knowledge regarding the research areas of the thesis project. See Table 1 for details regarding the respondents. These discussions were explorative and had only one main purpose respectively. With the Business developer, the purpose was to gather knowledge regarding how visions are created and handled within organizations. For the Expert engineer, the focus was to increase knowledge regarding the service market, workshops, and truck diagnostic tools. The findings of these discussions were used to design the interview guide as they provided knowledge and inishgt regarding vision development and the aftersales system.

Table 1: The respondents of the discussions with experts.

| Role | Area of expertise | Length |
|---|---|---|
| Business Developer | Technology strategy | 60 min |
| Expert Engineer | Diagnostic for service market | 60 min |

### 3.2.4 Field visits

The Field visits were conducted at two service workshops in the Stockholm area. See Table 2 for details regarding the field visits. The purpose of the field visits was to gather insight regarding how the current aftermarket system is designed and the workshop's role and responsibilities. The field visits also aimed to understand the workshop personnel’s viewpoint regarding how a driverless future would affect the workshops. This understanding was then utilized by the researcher when designing the interview guide. At Field Visit 1, the visit mainly consisted of discussions with different personnel and one customer and a walk-around through parts of the workshop. Field visit 2 at the second workshop mainly consisted of a walk-around combined with a continuous discussion with two customer recipients and an observation of a mock-up scene of a typical conversation between a service planner and a truck driver.

Table 2: Respondents from field visits.

| Field visit | Role |
|---|---|
| 1 | Workshop manager |
| 1 | Service advisor |
| 1 | Service advisor |
| 1 | Owner of a haulage company |
| 1 | Technician |
| 2 | Service advisor |
| 2 | Service advisor |

### 3.2.5 Semi-structured interviews

Semi-structured interviews were conducted in two rounds during the thesis project. Two different interview guides were used during the rounds (See Appendix A and B). As previously mentioned, the knowledge and data gathered from literature, discussions with experts, and field visits were used to construct the interview guide, especially for the interview guide for round one (See Appendix A). The interviews were between 45 and 90 minutes long; some were conducted face-to-face, while others were performed digitally. The semi-structured interviews focused on the respondent's vision of a service market system for driverless trucks and how the organization tackles transformations. All of the interviews were recorded and transcribed.

The respondents for the semi-structured interviews were carefully chosen based on their level and position in the organizations of Scania CV and Traton. The respondents consisted of people at three different levels in their positions: upper managers (Presidents of departments), lower managers (Presidents of sub-departments), and non-managers. The respondents were from a total of five different departments of Scania CV: Aftersales - R&D, Autonomous – R&D, Autonomous – Business, Connectivity - R&D, and Strategy - R&D. As well as one department from another brand within the Traton Group, Aftersales - Business. The departments are described briefly in Table 3.

Table 3. The different departments of the respondents for the semi-structured interviews.

| Department | Description |
|---|---|
| Aftersales – R&D | An R&D department responsible for the technical aspects of aftersales of trucks and busses at Scania CV. |
| Aftersale - Business | The department handles the business aspects of the aftersales activities and offerings for the brand in the Traton Group. |
| Autonomous – R&D | An R&D department that handles the development and technical aspects of the autonomous system on a Traton level. |
| Autonomous - Business | A department responsible for business development for autonomy, both confined and on-road applications, on a Traton level. |
| Connectivity – R&D | An R&D department that is working with the development of connectivity features for Scana CV’s trucks and buses. |

The main focus of the first round was to gather the visions of upper managers of the aftersales system for driverless trucks and learn how the organizations are working with the implementation of driverless trucks. The first round of interviews consisted of nine respondents, eight of whom were upper managers and one with no managerial role, see Table 4. The respondents were from both organizations, including Scania CV and the other Traton brand, as well as all the various departments.

Table 4: Respondents from the first round of interviews.

| Role | Department | Length |
|---|---|---|
| Upper Manager | Aftersales – R&D | 85 min |
| Upper Manager | Aftersales – R&D | 85 min |
| Upper Manager | Autonomous – R&D | 60 min |
| Upper Manager | Autonomous - Business | 85 min |
| Upper Manager | Connectivity – R&D | 50 min |
| Upper Manager | Strategy – R&D | 45 min |
| Upper Manager | Aftersales - Business | 90 min |
| Upper Manager | Aftersales - Business | 85 min |
| Project manager | Aftersales - Business | 85 min |

*The respondents were interviewed at the same time.

The purpose of the second round of interviews was to gather the visions from lower managers and people with no managerial roles regarding a service market system for driverless trucks, and collect more data on how the visions and expectations might vary within the organization of Scania CV. The second round of interviews consisted of five respondents. Three of them were lower managers, and two had no managerial roles. See Table 5. The respondents were from three departments: Aftersales - R&D, Autonomous - R&D, and Autonomous – Business. The interview guide was updated for the second round to target key areas that had been identified during the analysis of the first round of interviews where more data needed to be sourced, see Appendix B.

Table 5: Respondents from the second round of interviews.

| Role | Department | Length |
|---|---|---|
| Lower Manager | Aftersales – R&D | 50 min |
| Data Scientist | Aftersales – R&D | 50 min |
| Lower Manager | Autonomous – R&D | 50 min |
| Solution Manager | Autonomous – R&D | 50 min |
| Lower Manager | Autonomous - Business | 50 min |

### 3.2.6 Coding and analysis of the material

After the material from the semi-structured interviews from rounds one and two had been transcribed, it was coded and analyzed. The material from the discussions with experts and field visits was also compiled and used in the analysis. The following structure was used during the thesis project to code the empirical data:

- After every interview, review meetings were held, during which the respondents' greatest insights and overall impressions were noted. Discussions on how to improve for future interviews were also conducted.

- Each interview was coded separately as soon as it had been transcribed throughout the interview phase, primarily to avoid having a pile of material that had to be digested at the same time and to reduce the feeling of getting overwhelmed by data.
- Every interview transcription was read through and later dissected, where key insights and citations were noted down in Post-It notes. By doing this, a good understanding of every interview was guaranteed and made it easy to go back to the main findings from each of them.
- Post-Its were put together into clusters to create first- and second-order themes, which framed the findings and supported the analysis of the data.
- This process was followed for both interview rounds. See Figure 4 for details regarding the clusters from round one and Figure 5 for round two.

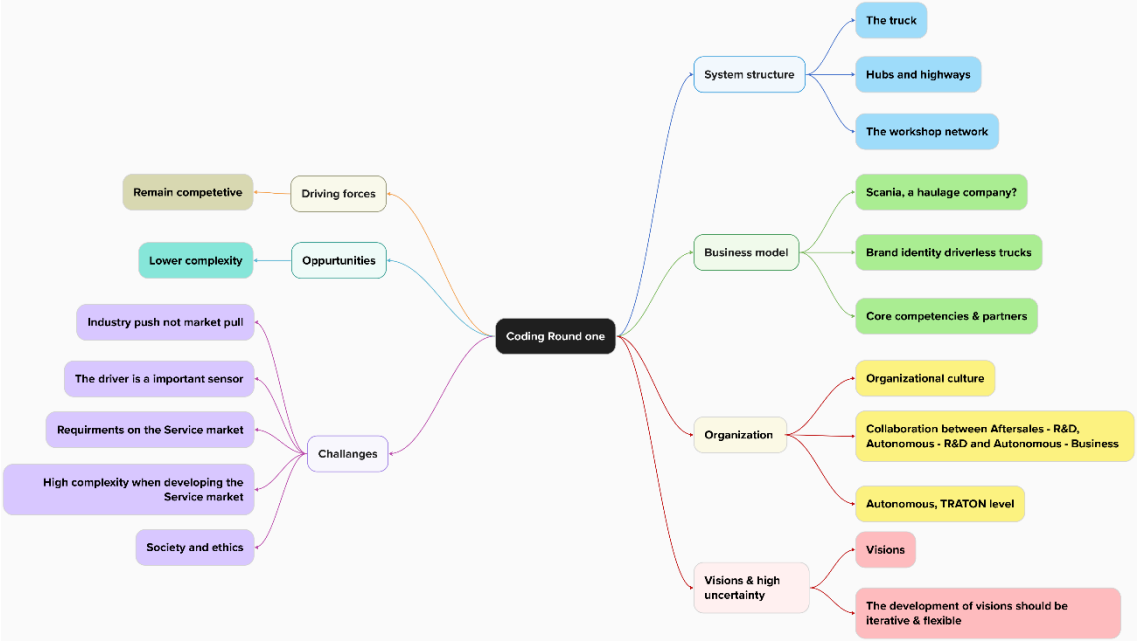



*Figure 4: Illustration of the first and second-order themes from interview round one.*

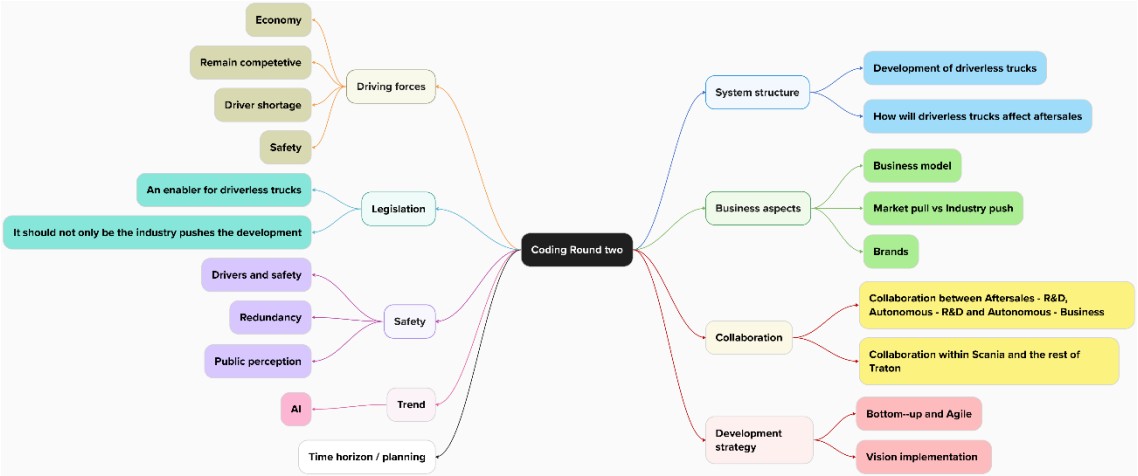



*Figure 5: Illustration of the first and second-order themes from interview round two.*

The following structure was used during this thesis project:

- All the material was compiled into an Excel document, where the different second-order themes got connected both within and outside of their first-order themes. The connections were discussed, and after that, the connections were added to the Excel file together with a score from 1-5 regarding relevance for the thesis project.
- The sub-themes with a score of 4 and 5 were seen as important findings and core findings, respectively. These were prioritized for further development with literature. The subtheme that scored a 3 was seen as more of a finding that could provide context and further depth to the ones with a greater score. The score of 2 was for findings that needed extra thought, and the researchers were unsure about presenting them in this report. The findings, which were graded with a 1, were not to be included in the results. The result of the analysis is presented in Figure 6.

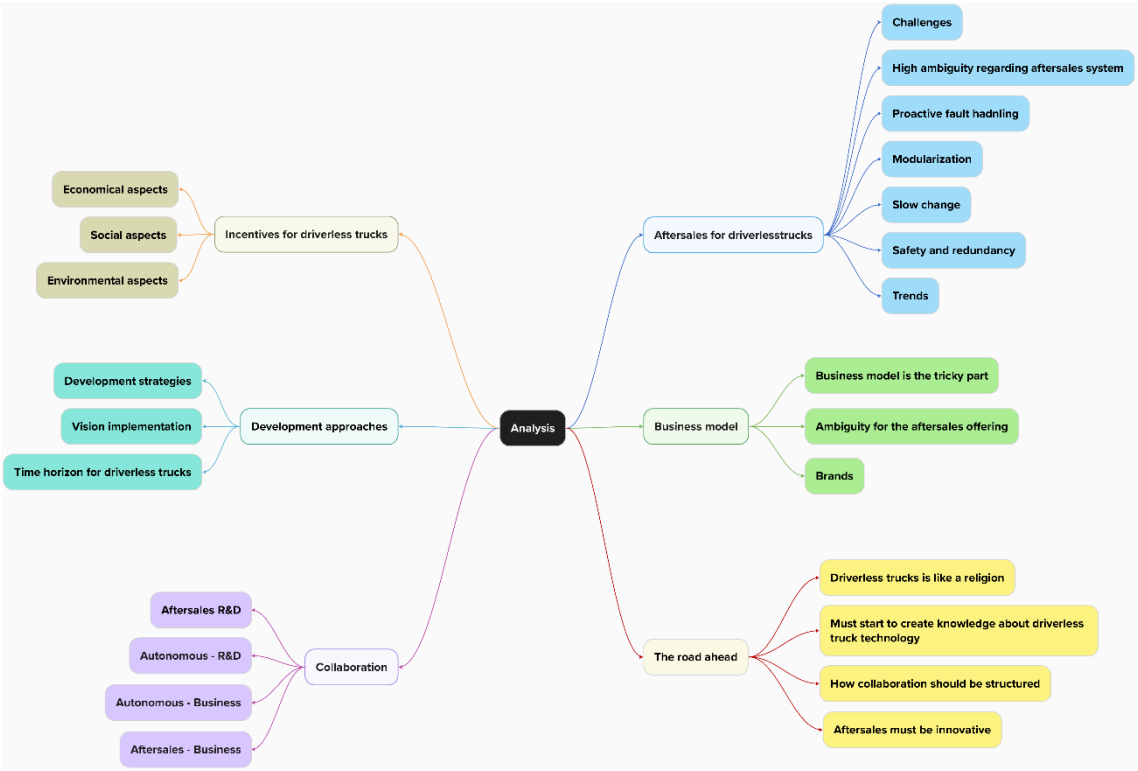



*Figure 6: Illustration of the first and second-order themes from the final analysis.*

# 4 RESULTS

*This chapter presents the results. It contains solely the empirical data acquired during the conversations with experts, the field visits, and the semi-structured interviews from the first and second rounds. Within the chapter, the references to the respondents vary. This decision was made to ensure a higher degree of anonymity of the respondents. Figure 7 shows an overview of the chapter.*




*Figure 7: An illustration of the sections presented in the results chapter.*

## *4.1 Development approaches and vision implementation at Scania CV and Traton*

In this section, the development approaches of the different departments from Scania CV and the other Traton brand will be presented. The findings focus on specific strategies for driverless truck development as well as general technology/business development for each respective department of the respondents. The section also examines whether visions are implemented as a tool for technology development and what time horizons are established in the organizations.

### 4.1.1 Development strategies for R&D and driverless trucks

As an organization, Scania CV is working with a bottom-up development strategy and is currently trying to establish agile development strategies to handle uncertainties better. Working with duality within the organization is important, being as agile as possible while also having customer demand as your key value and developing cost-effective products. Product development is an iterative task, and it is essential to focus on knowledge creation when uncertainty is high. However, customer awareness and short feedback loops are not as pronounced in the R&D departments today as they were previously.

According to one of the respondents from round one, the change rate for driverless truck technology is incredibly high, meaning it is important to be as open-minded and flexible as possible with your development strategy. To create a holistic and well-rounded offering, technology and business development must be conducted in parallel and in small increments.

The Autonomous – R&D, and Autonomous – Business departments that are responsible for driverless truck technology have a strong bottom-up culture where top-down management is regarded as dangerous, as it inhibits creativity. Multiple of the respondents from the Autonomous departments emphasized that it is important to be as agile and iterative as possible and create knowledge instead of focusing on pure results, as uncertainties are commonalities for driverless truck development. They also highlighted the importance of taking small steps instead of trying to develop the final product right from the start, frequently evaluating the development processes and current scope to make changes if needed. This is a tricky scenario as managers must always sense whether they should increase or decrease the freedom of their teams. One of the lower managers from round two expressed that it can be illustrated as a sinus wave that oscillates depending on the current grade of ambiguity in the development.

*"How much freedom do you want to give to your organization, and how much do you want to frame them? … I don't think that we still know the right answer yet"* – Lower manager, round 2

The Autonomous - R&D and Autonomous - Business departments function as start-ups within the organization, having the freedom to be flexible with their processes and remain as agile as possible. These departments are responsible for driving driverless truck technology and, therefore, need freedom to operate sufficiently.

*"The Autonomous departments should hold the baton and show the way forward"* – Respondent, round 1.

One of the respondents highlighted the perspective that these departments' primary focus is to develop a product that is as good as possible while neglecting cost aspects, which opposes Scania CV's traditional culture of providing both premium and cost-effective solutions.

For the Aftersales—R&D department, one of the upper managers reasons for the development structure being bottom-up. However, according to one of the lower managers, the department has lost some of its bottom-up characteristics. This respondent also states that there currently are many projects in the department, meaning there is limited room for new ideas and that futuristic projects have a lower priority.

### 4.1.2 Vision implementation

According to one of the experts, a vision is not a vision if it does not convey a collective picture of where the organization wants to be and must be shared predominantly by the actors within the organization. The respondent also states that it is common for upper and lower management to have different interpretations of the visions due to their different focus areas (strategy vs. technology).

From the perspective of one of the upper managers, vision implementation is important for product development as people need a clear direction of where the organization is moving. At Scania CV, high-level missions are supplied by Traton and then further broken down and divided into Scania CV's different departments. The desired future is a representation of where Scania CV wants to be as an organization as well as of what the customers demand. Strategy development for investments and budgets is handled by upper management and contains an assessment of multiple potential visions, where risk and opportunity are evaluated. This is later transitioned to the different subgroups of each department where upper management supports vision and strategy implementation among lower management.

The outlook of vision implementation varies within Scania CV's aftersales department. According to one of the upper managers, the department analyses Scania CV's high-level visions

and tries to break them down to understand how they are affected by them. Scania CV's main vision is to make the transportation industry more sustainable, and the clearest sign of what the organization wants to focus on is on what area resources are allocated to. However, according to one of the respondents from round two, Aftersales must be better at dissecting R&D's strategic plan, as the vision of the Aftersales department can diverge from it. The respondent also states that the aftersales department has more of a short-term perspective, which affects the implementation of visions.

One of the upper managers at the Autonomous departments states that visions play an important role in strategy development for the driverless truck development. The respondent also declares that the current vision has three branches, clearly describing their overall focus for driverless truck development. In addition, the respondent emphasizes how the product's lifecycle affects the vision scope, creating a need for more frequent vision refinements if there are high uncertainties regarding upcoming milestones.

*"The difficult thing is to be strong enough and stick to a vision so that you have something to hold on to when it starts to sway, but also mature enough to dare to say that the vision must change when it's time to get continued development."* – Upper manager, Autonomous department.

One of the lower managers declares that the subgroups of the Autonomous – R&D department usually get an abstract development strategy handed to them, upon which future short-term achievements are based. The strategy is mapped into "value creation trains" to make the strategy easier to interpret. The respondent also mentions that upper managers are often included in this process to guarantee that the correct milestones are targeted and defined correctly.

According to the Aftersales—Business department, vision implementation is an important tool that is tightly connected to the department's purpose. The vision should be easy to grasp and not too specific, and it should be continuously reevaluated. Two of the respondents from the department refer to their vision as a "northern star" and interpret it as guidance for the road ahead.

### 4.1.3 Time horizon for driverless trucks in hub-to-hub applications

When it comes to assumed time horizons for driverless trucks in hub-to-hub settings, the Autonomous – R&D and Autonomous - Business departments have a clear picture of what the future will look like. One upper and one lower manager states that there will be driverless trucks in commercial hub-to-hub operations in both the US and EU relatively soon. One of them also assumes that driverless trucks will account for a large part of long-haulage transport in the US in the near future. However, the upper manager mentions that the timeframe has a margin of error of a couple of years, while two lower managers from the Autonomous departments imply that the technology is too immature at this stage to make grounded statements. One of the lower managers also adds that it is difficult to make concrete plans/forecasts that go beyond three years into the future, as they become very speculative.

When investigating other departments' outlooks on driverless technology for hub-to-hub applications, the answers start to diverge. For instance, as for the aftersales—R&D departments, some of the corresponding respondents think that driverless trucks will be used for hub-to-hub transports relatively soon, while others believe that it will take a long time before we see driverless trucks on the roads in a commercial setting.

The same goes for the Aftersales – Business department, as there are various outlooks on the implementation rate for driverless trucks as well. One respondent believes that driverless technology will not be used for hub-to-hub applications for a long time, while another thinks that it will be introduced in this decade.

## 4.2 Collaboration within the organization

The following section will describe how the departments collaborate regarding the development of trucks. The findings will be divided into the perspectives of the relevant departments.

### 4.2.1 Aftersales - R&D

The Aftersales – R&D department has collaborations with the rest of Scania CV's R&D departments. They are involved early in the development process of projects. However, it is still too late, and they should be involved earlier as it would result in more complete solutions. If the Aftersales – R&D department had been involved earlier in projects, they would also have had better opportunities to implement their visions and set their directions in projects. Today, they rather work and develop based on the results of other departments. This could also lead to prospects regarding long-term vision implementation for the Aftersales – R&D department. One of the respondents from round two presents that the department has also struggled to get through with their opinion in the development of trucks, which historically has resulted in a lack of focus on aftersales functions, such as less appropriate designs regarding serviceability. However, they have the authority to block projects, even though "it is an uphill battle", due to that all R&D projects have to go through the department.

*"Aftermarket is like a huge funnel. I mean, all projects at Scania CV will need to go through aftermarket before they can be released, which means that the expertise sitting in aftermarket is gigantic."* – Respondent, round two.

One of the respondents from the Aftersales – R&D department argues that they have tried to be more involved in the driverless truck projects, but it has been difficult due to the early phase of the project. There are different reasons why the department has limited involvement in driverless truck development. One could be that the future of driverless trucks is still very uncertain, and therefore, there has been limited focus on the aftersales aspects. Another could be that the Aftersales – R&D department doesn't have driverless trucks in their scope yet, as their focus is elsewhere at the moment. However, it is important not to fall into the trap of not focusing on the aftersales aspects for too long. One of the upper managers argues that Scania CV has too much focus on optimizing the technology instead of also focusing on the aftersales aspects when working on development projects. It is important to develop the aftersales aspects proactively, especially when dealing with a great technology shift, which is what a driverless truck would represent.

*"The Aftersales department has tried to be more involved and has put more energy into it, but many of the vehicles being built are prototype vehicles, which makes it difficult to build something generic that can be used out there."* – Respondent, Aftersales - R&D.

### 4.2.2 Autonomous - R&D

Autonomous - R&D is one of the departments directly involved in the development of driverless trucks. Due to the pretty vague future of driverless trucks and the long-term scope of the Autonomous – R&D department, there are difficulties with collaborating with other departments at Traton. The upper manager at Autonomous – R&D means that they collaborate with the aftersales department as well, but that the collaboration could be improved. The respondent also argues that everyone who works within R&D needs to have aftersales aspects in mind when working on development projects. However, the lower manager at the Autonomous – R&D department has no experience collaborating with people from Aftersales and has no clear insight regarding the aftersales functions in general. The respondent also raises concerns that it feels like

the Aftersales department may not have the time to get involved in the driverless truck development before it is too late.

One of the respondents from round two describes the autonomous departments as isolated groups, almost like a start-up company within the organization. The respondent also highlights that this may be a conscious decision from the department to remain agile. However, the autonomous departments cannot stay in parallel with others for too long as it will be important to collaborate in the future.

### 4.2.3 Autonomous – Business

The collaboration between Autonomous – Business, and Aftersales – R&D consists only of discussions today, and due to that, they have not yet started to build the systems. However, it is soon time to start to think about the aftersales aspects, especially what the difference should be and is going to be between the aftersales of Scania CV and Traton. One of the respondents from round two argues that the collaboration between the department and others has improved, but it could be better. The respondent also emphasized that the department has a tight collaboration with Autonomous – R&D in the development of driverless trucks.

*"The product is immature, so the Autonomous – Business and Autonomous – R&D departments need to closely bounce ideas and concepts around and ensure that they are not far from each other when they get to the pitch."* – Respondent, round two.

### 4.2.4 Aftersales – Business

One of the respondents presented that the Aftersales – Business department has to adapt according to what the other departments decide upon and solely focus on offering as high uptime as possible to the customer. The respondent explains that the production department focuses on producing trucks as fast and as cost-effective as possible, which sometimes can be incompatible with high serviceability. It is an industry standard that the Aftersales departments are down-prioritized within the different companies. One of the upper managers at the Aftersales - Business department presents that there is a limited collaboration between them and the departments that develop driverless trucks. Another of the upper managers in the same department argues that driverless truck-related development is a low priority within the department.

## *4.3 Visions of the aftersales system for driverless trucks*

This section presents the various visions and expectations of the aftersales system for driverless trucks. Findings from the different departments are compared to generate a holistic image of the assumed future.

### 4.3.1 Challenges with fault handling for driverless trucks

The most significant driving force behind the development of driverless trucks is the lower TCO ownership for the customers. The driver itself is a great cost for the truck industry, covering approximately a third of the total cost. The truck itself would be more expensive if it were driverless. However, due to the possibility of more hours on the road, which the truck industry could utilize with a driverless solution, the truck industry is an appropriate business case for driverless technology. This is due to the time restrictions for drivers at the moment, having to take a break every 4,5 hours according to labor legislation. One of the respondents from round one presents that driverless trucks would enable the possibility of 24/7 transportation. However, some of the respondents from field visit one do not share a clear view of the driving forces for driverless trucks and instead emphasize that they do not see the need for having driverless trucks.

*"Actually, one of the most limiting factors is the time restriction on the drivers, they have to take breaks. This is the most restricting factor that is taken away when we are moving to an autonomous solution."* – Respondent, round one.

Some of the upper managers have the perspective that there is a great challenge in replacing the truck driver regarding fault handling, as the truck driver can be described as a sensor, one of the most important sensors of the whole system. The truck driver plays a great role in fault detection and fault handling of the system. Faults regarding the hardware parts are often detected by the driver and not by sensors. This is agreed upon by some of the respondents from the field visits, who describe that the input from the driver that the service workshops receive is important in fault detection, especially for hardware parts. Driverless trucks must have all the necessary capabilities to detect any kind of possible fault in order to be safe.

One of the upper managers raised another concern regarding fault detection for driverless trucks is that you cannot detect all faults with the fault codes today, questioning if it really is possible to rely solely on sensors for fault detection. The respondent also argues that today, some of the sensors and support systems of the trucks do not work correctly in some weather conditions, for example, in a blizzard. One of the lower managers presented a reasonable statement, the simple aspects of today's trucks, such as fault handling by the side of the roads, will be a great challenge for driverless ones.

*"When you remove the driver, you also lose a major sensor. A driver can hear if there are any abnormal noises, feel in the steering wheel if there are vibrations, or notice if the brakes are sticking. They can then report this to the transport company and the owner, who can plan a workshop visit to fix the problem."* – Upper manager.

One of the upper managers presented that a lot of drivers have great instincts. However, there are examples of drivers who continue on their route despite error messages occurring. Another upper manager explained that the trucks are becoming more complex, and this leads to the truck drivers sometimes not keeping up, which affects the fault handling capabilities. A respondent from round two argued that there is uncertainty regarding how truck drivers contribute to breakdowns. This opinion is also shared by some of the service workshop personnel from the field visits as they argue that truck drivers know less about their vehicles. There are also occasions when the driver does not support the fault detection process and instead misleads. However, an upper manager explained that some faults that occur today may not be relevant when the truck becomes driverless. For example, you could probably remove the windbreakers. Some of the actions that previously were done by the truck driver are now also managed with the help of sensors in the pre-check process before starting a route.

*"There may be a certain option that I can't think of where you have a hardware failure that has not been tracked by a sensor. For example, if your windscreen is broken, then this may be the case. But what do you need a windscreen for if you have an autonomous truck?"* – Upper manager.

There is also a safety aspect to consider when replacing the driver. A respondent from round two meant that some argue that a driverless truck would be safer than one with a driver due to tight driving schedules and the risk of bad working conditions, which can lead to a small room for the driver to take a break and rest. An upper manager presented an example, which is the game of chess. There are now robots that can win against human players, and in the context of trucks in traffic, approximately 90 percent of accidents are due to human error, there is room for improvement in this field regarding a driverless solution.

*"Many will probably disagree with me, but the computer will be a better driver than humans. Today it is not, but tomorrow it will be."* – Upper manager.

### 4.3.2 The demand for uptime will increase with driverless trucks

One of the major expected challenges of the implementation of driverless trucks is that faults will be more expensive. According to respondents from the Aftersales -R&D, it will be very important to have a service network of high quality with proactive fault handling, as it is the most important factor for providing high uptime. At the same time, driverless trucks can increase unscheduled downtime due to increased fault handling complexity as there is no driver present in the vehicle. Due to this, the Aftersales department will need to integrate more with the truck's sensors to communicate more with the vehicle and collect more data.

Another of the respondents from Aftersales – R&D also argues for increased expectations on the aftersales system as driverless trucks are starting to be used for long haulage transports. Since the trucks will be more expensive and have the opportunity of being operated more frequently, as no driver needs to rest, the aftersales system must be optimized to allow for the increased utilization rate. Additionally, the respondent declares that 24/7 transport is a possibility with driverless trucks and that the service network is an enabler for it.

The thought of service is even more important for driverless trucks compared to manually operated ones is shared by the Autonomous departments as well. However, one of the upper managers stated that trucks tend to get more and more complex overall, primarily due to the increased rate of electronics and embedded systems; the workshops have an increasing demand for them regardless of whether the vehicle is driverless or not.

Furthermore, there will be a need to monitor the uptime and performance of driverless trucks to make sure that the vehicles are up and running. A driverless truck must achieve the uptime levels that the customer has been guaranteed, creating a need for a monitoring function or control tower. A control tower with real-time monitoring would also stand for safety precautions, as a driverless truck that malfunctions on a highway can be a safety risk if not noticed by the operator.

The expectation for increased requirements on the service network is further supported by the Aftersales – Business department, as the service network performance will be a deciding factor for the success rate for driverless trucks. The respondents also reason for a shift in the service network footprint with driverless trucks as it will be easier to schedule for service since planned downtime is not restricted to a driver's working hours. It will also be easier to structure the logistics of spare parts and where a vehicle will be serviced, as the workshops will always know where the vehicle is. At the same, the service network will still need to handle regular trucks as well and should be as efficient and effective as possible, regardless of truck type.

### 4.3.3 Utilize the existing service network

To make the transition towards driverless trucks as smooth as possible, the existing service network should be utilized as much as possible. An interesting idea, expressed by one respondent from aftersales R&D, is also to add autonomous supervision/service halls to the existing service workshops to increase automation on a system level. Another respondent highlighted that it would take time before driverless trucks affect the industry. Therefore, the case could be that there are a lot of driverless trucks on the roads without requiring any big revisions of the workshop infrastructure. Scania CV also has an advantage over competitors, such as smaller start-ups and other companies, due to its existing service network, which can be utilized for driverless trucks.

According to the Autonomous – Business department, there will be a small difference between today and the future regarding service activities with driverless trucks. Therefore, the existing service network should be utilized, and the personnel should be educated regarding service on driverless trucks. However, the service function should be accompanied by a 24/7 monitoring system that applies to the entire organization of Traton.

### 4.3.4 The impact on the aftersales system and transport industry

The impact of driverless technology on the aftersales system and transport industry overall is rather ambiguous. For instance, if done sufficiently, driverless truck implementation can change the entire transportation industry. On the other hand, a respondent argued that the hype over driverless trucks has decreased recently, getting less attention than before. The respondent also suggests that the industry will urgently adopt driverless trucks as the technology matures. The complexity of the transportation system will also decrease since the operator will have more control of the vehicle compared to a physical driver. However, the complexity will primarily decrease for the customer, as some of their main responsibilities will transition to the manufacturer instead.

In contrast to the potential of revising the complete road transportation system, some respondents argue for a minor change as driverless trucks start to emerge on the roads. The introduction of driverless trucks will not be as revolutionary for the industry compared to what many want to believe. Also, driverless vehicles should only be seen as a part of the transportation ecosystem, not replacing other key functions where manually operated trucks are a better fit.

Accordingly, one of the respondents from round two expresses how driverless trucks will be introduced to the market in increments, and the industry's required adoption rate will, therefore, be slow. For instance, in hub-to-hub use cases, some transport might be conducted without a driver in the vehicle, while some road sections might require a physical driver. Therefore, there is no need for a fully functioning system optimized for driverless vehicles, just as they are starting to get commercialized.

*“The realism has caught up to us.”* – Respondent, round two.

*"It's not something that will be turned upside down overnight.”* – Respondent, round two.

The same respondent presented that for the same reason, the required adaptions of aftersales functions and the service network will be limited for driverless trucks as they start to emerge on the roads. The only challenge might be that workshops will require new competence to handle those types of vehicles.

*"It sets a slightly different set of requirements. But that's where you have room to adapt to the service market. You know that something new is emerging that might require a slightly different skill set."* – Respondent, round two.

One of the respondents from the Aftersales - R&D department is more speculative when it comes to the technological aspects of fault detection for driverless trucks and believes that trucks must communicate and thereby could diagnose each other with cameras. This can be taken to a wider extent where different manufacturers share data to allow for diagnosis between brands. The respondent also highlights that the departments that are responsible for driverless technology development often have an optimistic approach toward driverless trucks and fault management, as AI is often regarded as the solution to everything. However, the respondent thinks that fault handling will be a more complex challenge.

From the Autonomous – Business department's outlook, aftersales for driverless trucks can be divided into two areas. Firstly, there is software-related maintenance, where the ”digital driver”

is getting software updates and improved hardware to allow for enhancements of the driverless technology. According to the respondents at the department, this is an area where there will be a need for immense work as driverless trucks are starting to be commercialized, as there most likely will be a great number of software updates at that stage. Also, within this service area, it might be beneficial to ensure that both hardware and software are compatible to increase circularity. However, the technology is yet too immature to start thinking about these aspects of the offering, according to the lower manager.

*"The solutions we have today will not be commercially viable in 3 years because both computer power and sensors and everything are developing so rapidly."* – Respondent, Autonomous - Business.

Secondly, there is regular hardware-related service, consisting of the typical component-based faults for conventional trucks According to the respondents, this type of service effort will not be affected by the transition to driverless trucks, as the driverless attributes will have minor effects on the rest of the truck's system platform. On the other hand, one of them emphasizes that the shift towards driverless trucks will affect the service network by requiring new competence among workshop personnel to handle the vehicles.

*"It will be a huge change for the service network, of course. It's like going from an internal combustion engine to an electric vehicle. In this case, it's almost as if data engineers will need to manage the service network. Currently, the system is very complex, and we need to look at how it's structured so that it's easy to calibrate and troubleshoot, for example."* – Respondent, Autonomous - Business.

From a service perspective, driverless trucks of autonomy level four will decrease complexity as the operator will always have control over the vehicle and know where it is going or what it is doing. It is also important to understand how the service network can be optimized for driverless trucks as the technology is starting to mature. For instance, the current service network might be suitable for smaller customers, while it might be an idea to establish hub-based workshops for bigger customers with larger fleets, which was presented by some respondents. However, one respondent from Autonomous - Business argues that hub-based workshops can be complex to establish. In addition, one respondent from Autonomous – R&D argues that hub-based workshops will not be a significant differentiator for driverless trucks and perceives that driverless trucks will still keep the human interface with a cab and steering wheel, making it possible to drive them to designated places or workshops if needed.

Furthermore, a high-quality service network is needed as high uptime will be the main objective for driverless trucks, although it is important regardless of the truck type and whether it is driverless or not. An upper manager also highlights the need for guaranteed maintenance within a set time or distance interval in the case of unforeseen failure to limit downtime. To avoid unforeseen faults and to understand what has failed in the event of a fault, it will also be important to collect as much data as possible from the vehicle.

*"If you have an autonomous vehicle, the vehicle needs to talk to you as clearly as possible, and we need to understand what is wrong."* – Upper manager.

However, with immense data collection, high demands will be put on the capabilities to analyze the data effectively and efficiently, something that can be a complex task. This statement was also further developed by expressing a change of complexity, where conventional customers will have less responsibility for fault detection as the manufacturer or operator will monitor the vehicle status.

Another thought, presented by an upper manager, is common diagnostics among Traton brands, as the number of available workshops for each Traton brand will increase, thereby positively affecting service capabilities. At the same time, it is important to maintain the relationship with

local workshops and ensure that they are not left behind as new technology is being introduced to guarantee a dense workshop network.

*"... smaller family-owned businesses will not be able to invest millions into technology to be able to keep up with the changes within the industry ... will we be perceived as a premium brand or not, if we don't have the same amount of people and the same amount of workshops in our network any more."* – Upper manager.

To ensure that knowledge is transferred to workshop personnel, it is important to utilize the academy functions at the Traton brands to ensure that mechanics have the right competence to take care of the driverless fleet. By pushing knowledge out to the workshops, driverless technology will not be as foreign to them as the first vehicles that need maintenance.

### 4.3.5 Proactive fault handling

According to Aftersales - R&D, proactive fault handling will probably be more important in the future with driverless trucks as there is a need to improve the capabilities regarding the life span of the different components and systems. A lot of resources are also devoted to this area today as preventive service is important to create an effective system with high uptime for all types of trucks, and the only demand that will change regarding aftersales for driverless trucks is that it will require even more proactive fault handling.

The service itself will look the same, with the difference of not having a truck driver who can explain how the truck has acted and the sequence of events. There is a lot of data to collect from the driver today. However, if the system were more data-driven by sensors and software, there would be more control and supervision of the truck, which would lower the complexity. More information is needed when no truck driver can convey any information when you are dealing with a system that needs more proactive fault handling.

*"We must become more skilled at predicting the lifespan of various components depending on how they are used. I believe that could be a clear requirement"* – Respondent, Aftersales – R&D.

### 4.3.6 Safety and redundancy

According to a couple of upper managers, driverless trucks will probably need more connectivity capabilities, making them more vulnerable to cyber-attacks. The respondent also explains that with a driverless truck, you lose redundancy as no driver can manage errors due to cyber-attacks.

*"I believe that cybersecurity is such a thing that will have to be, it must be 100% secure, not 99.9%, it must be 100% secure, otherwise autonomous vehicles will not make a breakthrough."* – Upper manager.

However, the system does not need to be designed to have a constant connection, only when it is desired and data must be transmitted. The most important aspect is having redundancy concerning connectivity, making sure that it is always possible to get in contact with the truck if there is a need to do so. Today, it is possible to remotely collect data from trucks, which is a valuable function for the workshops that are dependent on connectivity The service workshops use this function to support the truck drivers. However, on occasion, it takes a lot of time to do a remote reading, and there is room for improvement to make it more consistent.

*"No, the truck does not need to be connected all the time... It's reverse logic. When the truck needs connectivity, it can cost just about anything."* – Respondent, round one.

One of the upper managers from the Autonomous departments stated that there is a good case for increasing redundancies for driverless trucks as there will be less acceptance of failures and faults. Redundancy is also important to avoid dangerous situations. Another respondent from the Autonomous departments also explained that the system needs to have knowledge and diagnosis tools for different situations, and there must be systems that ensure that the driverless truck can drive to a safe place if anything goes wrong.

*"Yes, but it will be completely unacceptable to have trucks on a standstill on Essingeleden haphazardly."* – Upper manager, Autonomous departments.

*"One must ensure that there is as much diagnosis as possible for what is dangerous, and otherwise, that there is a very secure way to maneuver to a safe place, and then you come to the fallback solution for the robot as well, that it must be aware and receive notifications that yes, wait now, it seems that something is not quite right, then it drives off the highway and goes to a safe place."* – Respondent, Autonomous departments,

### 4.3.7 Modularization as a strategy for driverless trucks

Scania CV has an extensive and well-developed modularization strategy for its trucks and has the capability to adjust the truck design based on the requirements of many different customers. The modularization strategy enables a great set of combinations of different technical solutions. The autonomous system package should be viewed as a part of the building block as it makes it easier to implement the driverless truck technology in Scania CV's current platform and, thereby, less complex regarding fault handling for the existing service network and existing functions for aftersales. One of the upper managers explains that the autonomous system was historically viewed as a separate product from the modular system. However, it is now viewed as a part of the modular system.

"*Autonomy is just a building block in the modular system*." – Upper manager.

Furthermore, the respondent argues that driverless trucks could be standardized based on the decided routes they operate on and that different variants can be used on different routes. This would be less complex for service workshops, as they would only have to focus on a standardized truck within the modular system.

Three of the upper managers also argue that the modular strategy enables better serviceability. One of them explains that from an aftersales perspective, the trucks should be as modular as possible to make maintenance more efficient and reduce the truck's downtime. However, there is a downside since a modular system can be less sustainable from an environmental perspective as complete modules are replaced and later scraped as it can be more difficult and resource-consuming to change a separate component within the modular system. An idea presented by two of the upper managers would be for the service workshops to swap the whole module that faults the truck, replacing it immediately with a functioning one, then letting the truck drive off and continue its operations while the service workshop continues to fix the fault on the broken module.

### 4.3.8 Trends

AI is a clear trend among the Autonomous and Aftersales departments and is spoken of frequently today as it can be used to analyze large amounts of data to recognize patterns that could enable higher uptime by enhancing the capabilities of proactive fault handling. AI is also a fundamental condition for driverless trucks concerning the driving itself and fault handling. Aftersales—R&D also utilizes AI today to support service workshops when conducting service on trucks.

Multiple respondents also mentioned AI as a trend and said that it is important for driverless trucks. One of the respondents from Autonomous – R&D also explains that the utilization of generative AI will increase as upcoming generations of driverless trucks are developed and released. Generative AI will make the validation work within the development process less resource-consuming. It is expensive and resource-consuming to scale driverless trucks today, but the development will be faster when there is an effective way of testing and validating concepts, like a "ketchup effect".

*"Yes, the AI trend is very clear."* – Respondent, Autonomous - R&D.

## *4.4 Business-related challenges*

This section presents findings regarding whether the development of driverless trucks is due to an industry push or a market pull. It also discusses the business model for driverless vehicles and aftersales. Due to a great deal of ambiguity, developing the business aspects of aftersales for driverless trucks is challenging. There is also a challenge regarding how to deal with brands and driverless vehicles.

### 4.4.1 Industry push vs. market pull

A great driving force regarding the development of driverless trucks is for truck manufacturers to stay competitive and continue to be relevant in the future. Scania CV wants to continue developing new technology to keep market shares and not take the risk of ending up being a hardware supplier for others. This is because the development of driverless trucks can be described as an industry push. This is a challenge as there is no market pull at the moment, and the customers are not willing to pay for the existing solutions. One of the respondents from Aftersales – R&D explains that there is a need to prove to the customer that the driverless truck solutions will work to get them to invest. Another respondent from the same department highlights that the new driverless technology must fulfill a purpose for the customer. One possible solution to handle this challenge is to engage in partnerships with other companies that are like-minded to create a stronger and unified push.

*" The solutions available now are not what customers are prepared to pay for, so this will take a few years longer than we thought."* – Respondent, Aftersales – R&D.

However, the perception of market pull for driverless trucks varies throughout the organization. Two of the respondents working in the Autonomous - R&D department argue that the demand for cheaper transport is a market pull for driverless truck development. One of them also emphasizes that all industries want to make repetitive work autonomous and that it is only a question of maturity, as automation usually implies cost savings in the long run. One of the respondents from the Autonomous – Business department argues that it was too much of an industry push earlier but that they have started to integrate customers in their projects regarding driverless trucks in a hub-to-hub context.

*"I rather think that there is a real pull from the customer regarding driverless trucks and that there is not always an understanding from the customer side of why the development takes so much time."* – Respondent, Autonomous – R&D.

### 4.4.2 Business aspects of driverless trucks

There is a lot of ambiguity regarding how the business model will be designed for driverless trucks. Scania should at least be responsible for the development, production, and service of

driverless trucks, but there is ambiguity regarding Scania CV's position in the value chain in the context of driverless trucks, according to one of the upper managers. Another of the upper managers presented that the customer interface is another complex aspect with the offering of driverless trucks and it is unclear to what extent Scania CV should integrate or collaborate with their customers. Legislation is also a great challenge for driverless trucks, as it affects business cases, and there is a need for new legislation that allows the usage of driverless trucks, according to one of the respondents from round two.

Another respondent from round two agrees that there is great ambiguity and uncertainty regarding the business model for driverless trucks. The product is not mature enough, and neither the business model nor the legal requirements are sufficiently defined yet. One of the upper managers argues that the most important aspect of the business model is to deliver a driverless solution and offer high uptime with it. Whether it is a system solution or a transfer of competence has not yet been decided.

A driverless truck offering will not be suitable for all business cases. One of the respondents from Autonomous – Business presented that a hub-to-hub business case that seems promising is where you have large truck fleets. The respondent also highlights a concern regarding the business model for driverless trucks today as Scania CV offers high value to the customer in the form of low fuel consumption and a great driving experience for the truck driver. Once the trucks have an electrical drive train and a system that allows for autonomous driving, all of that is lost. The other respondent from Autonomous – Business argues that whoever integrates themselves best into the customer's business will be the market leader. The respondent also describes the business case as rather clear, and has a comprehensive perception of how the offering should be structured towards the customer. The other respondent from the same department explains that the customer should be responsible for all logistics and that Scania CV should focus on optimizing the autonomous driver. Both of the managers argue that the US is a great business case for driverless trucks in a hub-to-hub context, as the legislation is no barrier to the development of driverless trucks. However, the liability aspect regarding how to divide the responsibility if anything goes wrong is a great challenge in the US.

Perspectives from other departments were the following. A great challenge regarding the future business model for driverless trucks is that the existing business model has not had any major changes in a long time, which will be a barrier. There are uncertainties regarding the business case for hub-to-hub, which will be the first industry and customer group for on-road autonomy. One of the upper managers clearly sees a first customer. However, the exact size of the fleet and how the offering should be designed for the customer is uncertain. Scania CV should just go in the direction that generates the greatest profit. the respondent also argues that it is important that Scania CV keep some capabilities within the company and that a ground rule should be to identify what Scania CV's strongest capabilities are today and investigate how these could be utilized when developing and offering driverless trucks.

*"You should just follow the money."* – Upper manager.

Two of the respondents from Aftersales - Business explained how Scania CV and Traton could expand their position in the value chain and offer transport operations with driverless trucks. However, they also stated that starting to compete with your customers is not optimal since they are highly competent in their area, and it would be bad for business.

### 4.4.3 Ambiguity with the business model for the aftersales offering

There is high ambiguity regarding how the business model of the aftersales offering should be structured. The main differentiator with driverless trucks, compared to the current aftersales

business model, is that revenue cannot be generated by vehicle malfunction to the same extent as before. Since driverless trucks should be up and running as much as possible to benefit from the increased utilization possibilities, bigger efforts must be put into preventive maintenance to make sure components are replaced before the vehicle breaks down. One of the respondents from round one declared how aftersales will be more integrated into the driverless offering, as the trucks will be offered as a service rather than a product, with a guaranteed utilization rate.

The respondents from the Aftersales—Business department have a clear outlook on the uptime offering. They think of driverless trucks as an opportunity to provide higher uptime to their customers due to the possibility of 24/7 operation. However, one of the respondents acknowledges that it is hard to foresee how the business model for the aftersales offering for driverless trucks should be structured.

One of the respondents from Aftersales agrees with the reasoning above and expresses that fully understanding the aftersales offering is complicated and that the organization is not there yet. The aftersales offering will need to be adapted for driverless trucks, but it is hard to foresee how since the business case is too ambiguous at the moment. However, the technical functionality to conduct preventive service will most likely be available in the vehicles by design, while understanding how the business side of preventive maintenance should be developed is the tricky part.

### 4.4.4 Brand identity's importance for driverless trucks and aftersales activities

There are assumptions that brand identity will be less important for driverless trucks. For instance, Respondents from the Aftersales department describe how important brand differentiators, like driver comfort, will be insignificant for driverless trucks and that the vehicles will become a commodity. At the same time, Traton brands collaborate more and more today, increasing the commonalities between the brands.

From the outlook of the Autonomous departments, driverless trucks primarily make transport more effective, and there is no incentive to offer various qualities of a digital driver to differentiate Traton brands from each other. Brands will most likely be used as driverless trucks are introduced to the market, even though brand identity and traditional differentiators, such as fuel consumption, are less important for driverless vehicles. The main selling point will instead be uptime, but the brand identity question is complex.

*"…it's really just a transport platform that should be measured purely on efficiency…"* – Respondent, Autonomous department.

The end product (driverless technology) will probably be the same regardless of the brand, while there might be differences between their respective sales organization as each brand knows its customers best and how the offering should be structured. The customers are most likely not familiar with Traton either, which makes it difficult to approach them without the brands and their respective brand cultures.

Another idea could be to rebrand existing workshops and make them Traton workshops, but it is unclear whether that is the right path to take. On the other hand, if it is beneficial to let go of brand culture, it should be done. The aftersales offerings are the biggest differentiator between the Traton brands, and aftersales can therefore be utilized to give brand identity relevance to driverless trucks

The respondents from the Aftersales – Business department agree that brand identity is less important for driverless trucks and that the total cost of ownership is the most crucial aspect. One of the respondents also implies that aftersales is the differentiating factor for a driverless truck,

while the vehicle itself is more of a commodity. The two Traton brands belong to the top tier regarding aftersales in Europe, which is an asset for providing a solid service function for driverless trucks and potentially even as a Traton service. However, integrating two brands with completely different cultures can be a challenge.

## 4.5 The road ahead for the aftersales system for driverless trucks

This section presents findings that show that there are a lot of different visions within the organization and that the Aftersale – R&D department should be involved in the development of driverless trucks. It also presents suggestions on how the departments should collaborate.

### 4.5.1 Many different visions regarding driverless trucks

There are a lot of strong visions and opinions regarding driverless trucks within Scania CV and Traton. One of the upper managers explained that the visions are more comparable to religion than facts. The opinions are regarding how much the driverless technology will affect the industry when it is commercial and in what way it will be profitable. This is due to the complexity of the field of driverless trucks. The respondent also further explains that it is difficult to form strategies around matters like this as the different departments and people have different requirements and aspects that they value highly, and therefore, there will always be someone who feels left behind.

Another upper manager explained that the different vision regarding driverless trucks could be due to the design of the organizational structure. It is designed as if there were several smaller companies with standardized interfaces between them. An additional explanation for why there are different visions within the organization presented by another upper manager is that outdated organizational visions that once were presented for the organizations continue to be spread, and the departments that do not focus on driverless trucks then still believe that it is the vision. This is the case for a vision that was developed regarding whether Scania CV should change its position in the value chain or not. The respondent also argues that the previous visions for driverless trucks have been too extensive and that you must dare to narrow them down.

*"It almost becomes more like religions in terms of what people believe rather than facts."* – Upper manager.

*"One must always consider many different aspects, and there is always someone's sick grandmother who has to be deprioritized."* – Upper manager.

### 4.5.2 Aftersales must be involved in the development of driverless trucks

One of the respondents from Autonomous R&D expressed that it is time for Aftersales - R&D to start gathering knowledge regarding driverless trucks, and there is a need to start collaborating with them soon; otherwise, it will be too late. Aftersales should get involved when there is a well-defined concept of the driverless truck for hub-to-hub applications, and to avoid missing important steps, the departments need to always be ready and understand what has to be done when new needs arise. It is important to collaborate concerning how fault handling should be managed without a truck driver, but also regarding the business aspect of aftersales and driverless trucks. as there is a need for technical competence when dealing with business development.

*"It's gonna be too late if they want to wait until we have a product."* – Respondent, round two.

One of the upper managers argues that specific data is required to be more sure of how aftersales should be adapted to driverless trucks, as there is a need for multiple driverless trucks on the road to acquire this kind of data. However, one of the respondents from Autonomous – R&D explained that they conduct fault analysis on their driverless truck system themselves to investigate what sensors are needed and how to incorporate redundancy. Another respondent means that there are a lot of similarities between today's trucks and future driverless trucks as the driverless ones will have common components, such as the brakes for example. Therefore, the tools and services that Aftersales – R&D develops today could be utilized in the future for driverless trucks.

### 4.5.3 Suggestions regarding collaboration

One of the greatest barriers to collaboration at Scania CV is the various focus areas of the different departments, making it difficult to prioritize similar requirements and understand each other. To improve the cross-functional collaboration at Scania CV and Traton for driverless truck development, exchange programs should be implemented where key personnel get the opportunity to rotate at different departments to create new knowledge, as proposed by one of the respondents from the Autonomous - R&D department. However, the respondent also argues for the importance of treating customers as partners to allow for external collaboration to build more knowledge that can benefit driverless truck development.

One of the respondents from the Aftersales – R&D department expressed that even though aftersales often are disregarded by other departments, there is a general interest among other R&D departments at Scania CV to understand how aftersales function and what they are working with. According to the respondent, the Aftersales – R&D department has held workshops previously for other departments to inform the rest of the organization about aftersales and its function. By continuing with this, the importance of aftersales can be spread even further across the organization. The respondent also suggested that collaboration among Traton's various aftersales departments should increase to make use of each other's resources, data, and competencies to improve the aftersales functions.

One of the respondents from the Autonomous – R&D department suggested how to gather data regarding fault handling from one of the Aftersales – R&D functions. Another respondent from the same department approved of this idea and argued that it would be a great way of collecting data. The respondent also presented that it is possible to gather data regarding aftersales activities from the business area "Confined" that can be used for hub-to-hub. However, there are fundamental differences between the two business areas. Therefore, it would be very appropriate to make use of the Aftersales – R&D function proposed to further explore the needs for driverless trucks from an aftersales perspective.

### 4.5.4 Need for aftersales to have a more innovative approach

Two of the respondents presented that to make sure that aftersales functions are developed sufficiently for driverless trucks in the future, it is important to take a step back to fully understand how the offering should be structured and not to rely on old processes, both for business and R&D development.

However, other respondents presented that it is impossible to do everything in advance, as the future is too ambiguous. Therefore, aftersales must try to analyze what the future holds and make incremental changes. Otherwise, according to a respondent from Aftersales—R&D, there is a potential risk of losing market share.

*"It's not about making a copy-paste of the old, but it's about really evaluating, analyzing, and looking at the new"* – Respondent, Aftersales – R&D.

# 5 DISCUSSION & ANALYSIS

*This chapter will discuss and analyze the empirical data presented in the Results chapter in combination with the theory presented in the Frame of Reference chapter.*

## 5.1 Visions and ambiguity

In this section, material from the Frame of Reference chapter and the Results chapter will be used to answer research question 1 - *What are the current visions of the aftersales system for driverless trucks within the truck industry/Scania CV, and what challenges can be foreseen in driving the development of the system?*

### 5.1.1 Introduction to the visions within the organization

Both the literature and the empirical findings show that visions are important for organizations. Visions provide a story containing goals and an aim (Bemler, 2023) and set a direction that motivates the organization to move toward a desirable future (Kotter, 1995). It is also important that the vision is mutually shared within the firm (Haque et al., 2016); otherwise, the involved actors within the organization risk moving in different directions. Some of the respondents agree with this, as it is stated that a clear direction is needed for employees who work with product development. Some of the respondents described visions as a "northern star," which clearly and brightly guides what direction the organization should take. Another respondent also mentioned that a vision needs to be shared within the organization, as otherwise, it is not a vision. This highlights the importance of a shared vision to avoid parties within the organization taking different directions.

Tightly connected to the vision and the direction of the organization are technological paradigms and technological trajectories, as the direction of the organization and the technological trajectories are affected by the existing technical paradigms (Dosi, 1982). Driverless trucks can be defined as a discontinuous innovation, as it can be argued that the shift to driverless trucks is a significant change in the truck industry. Therefore, the transition towards driverless trucks forms a new technological paradigm. Hence, the technological trajectories and visions of the organization are changed.

### 5.1.2 Mutual and shared visions within the organization

There are multiple visions within the organization of Scania CV regarding how the aftersales system will be affected by the introduction of driverless trucks, see Figure 8. Even though driverless technology is a fast-developing field with high ambiguity, the various departments that were involved in this thesis have similar outlooks on many objectives, challenges, and requirements related to driverless trucks and aftersales development. To start with, a majority of the respondents agreed that uptime is one of the most central aspects of driverless trucks and that the aftersales system is one of the main enablers for providing high uptime. The findings from the interview study indicate that demands for availability on driverless trucks are assumed to increase compared to manually operated ones. This is mainly linked to the fact that driverless trucks can be operated on a 24/7 basis, which puts high demands on aftersales functions to enable a high utilization rate.

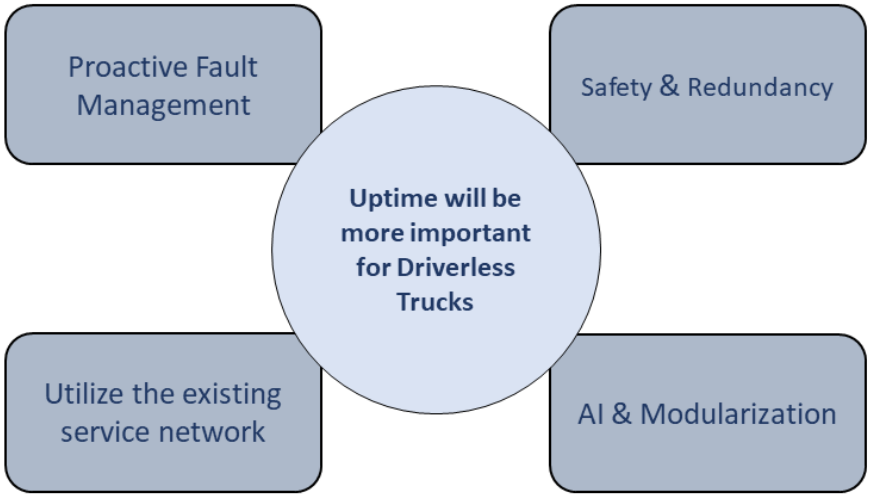



*Figure 8: Illustration of the shared visions of the aftersales system for driverless trucks.*

According to some respondents, one of the primary necessities for enabling high uptime is proactive fault management, as maintenance only is conducted when it truly is needed, which can limit unscheduled downtime while also making service more cost-effective and sustainable as resources are not wasted on unnecessary maintenance. This goes hand in hand with the reasoning of Allmendinger & Lombreglia (2005) and Porter & Heppelmann (2021), as the incorporation of smart products and smart services allows for enhancements of service and maintenance activities, as is it possible to utilize proactive maintenance due to the enhanced connectivity and components.

Other respondents highlighted redundancy and safety as important aspects for aftersales-related aspects of driverless trucks. The respondents perceive the acceptance for driverless truck malfunctions to be lower compared to manually operated ones, which will require the trucks and aftersales functions to have well-designed safety protocols to handle unexpected errors, both to make sure that unscheduled downtime is minimized but also to eliminate road-related safety risks. Here, multiple respondents emphasized how AI is seen as an enabler for the future development of driverless technology, both since it currently is implemented for fault detection management and is perceived as an important tool for data analysis and monitoring of driverless trucks, but also since regenerative AI can be implemented in training the embedded systems to develop a mature product quicker.

Furthermore, the findings from the interviews show that driverless technology should be implemented into Scania CV's modular system and that the current service network should be utilized for driverless truck maintenance to as wide of an extent as possible. By making driverless technology into a module, it will be easier to integrate it into the current product platform, which simplifies the transition for the aftersales department and service network as there will be more similarities to Scania CVs' current products, compared to if driverless trucks would be designed as a new type of vehicle from the ground up. This strategy also makes utilization of the current workshop network possible, as the virtual difference compared to a conventional truck will be rather small from a maintenance perspective, as there will be no mechanical difference between the two truck types.

One of the respondents also pointed out that being able to utilize the current workshop network is a great asset when introducing driverless trucks commercially, as it is a foundational

requirement to be able to guarantee high uptime. According to the respondent, Scania CV's service network is very thorough and is also one of the organization's main focus areas and another reason for its success. As an incumbent in the road transport industry, this capability should be seen as an advantage compared to start-up organizations that do not retain the same resources in terms of service or support capacity.

Consensus among certain aspects of the future for driverless truck development can be a result of clear visions where organizational culture is well established: modularization and the potential for revenue in aftersales offerings. Even though some respondents from the aftersales departments highlighted that aftersales are disregarded to some extent in development projects, there is consensus about the importance of aftersales for driverless trucks, as uptime is the primary selling point for that type of vehicle. At the same time, requirements for uptime and proactive fault management are being prioritized today, and they are highly relevant for conventional trucks, therefore making it easy to transition that vision to driverless trucks as well. The same goes for modularization and AI, as these themes are highly relevant to Scania CVs' current operation.

According to (O'Reilly III & Tushman, 2004), the success of an organization where ambidexterity is used as a development strategy is partly dependent on clear vision communication as it creates a foundation for a homogenous understanding of where the organization is moving among its various sub-departments. Accordingly, it seems that the most central strategies/values of Scania CV are shared by the different departments that have been included in this study, showcasing how there are shared outlooks on the aftersales-related attributes of driverless trucks between the aftersales and autonomous departments.

On the other hand, as emphasized in (Lewis and Clark, 2020), subgroups in large organizations tend to develop individual visions within their respective division, either resulting in a negative state where the visions are contradictory to other departments or the organizational vision, or a positive outcome where the different visions complement each other. This could be the case for the outlook on the aftersales-related aspects of driverless trucks, as the most fundamental concepts of Scania CVs strategy are shared by the different departments, while there are different nuances in what is perceived as important for obtaining the strategic objectives. For instance, even if most respondents perceived uptime to be important for driverless trucks, analysis of the findings from the empirical data indicates that the importance of adopting the modularization strategy and that driverless technology should be implemented as a building block primarily was highlighted by top managers. Also, the importance of proactive fault management was mostly emphasized by the aftersales department, while the other departments primarily focused on safety and redundancy. This shows that even if a majority of the respondents agreed upon uptime being one of the most important factors for driverless trucks, nuances in their perception vary and focus on different aspects of enabling high uptime by pointing out different components that must be taken into consideration for the aftersales system to be structured sufficiently.

Here, synergies to the properties of sensemaking can be identified, more specifically, the retrospective property, which suggests how people base assumptions of the future on previous knowledge (Helms Mills et al., 2010). This means that the respondents sense or attempt to make sense of the aftersales system for driverless trucks by basing their perception on their previous experience regarding what they assume is important for aftersales activities. As mentioned, retrospective is an important part of sensemaking (Weick, 1995), which can explain why there is consensus on the importance of uptime in general but various nuances of how uptime can be promoted.

### 5.1.3 Vision within the organization that is affected by ambiguity

This subchapter is divided into three parts, "*Technology-related ambiguity", "Business-related ambiguity*," and *"Different visions between departments and hierarchical levels"*. See Figure 9 for a summary of the ambiguities.

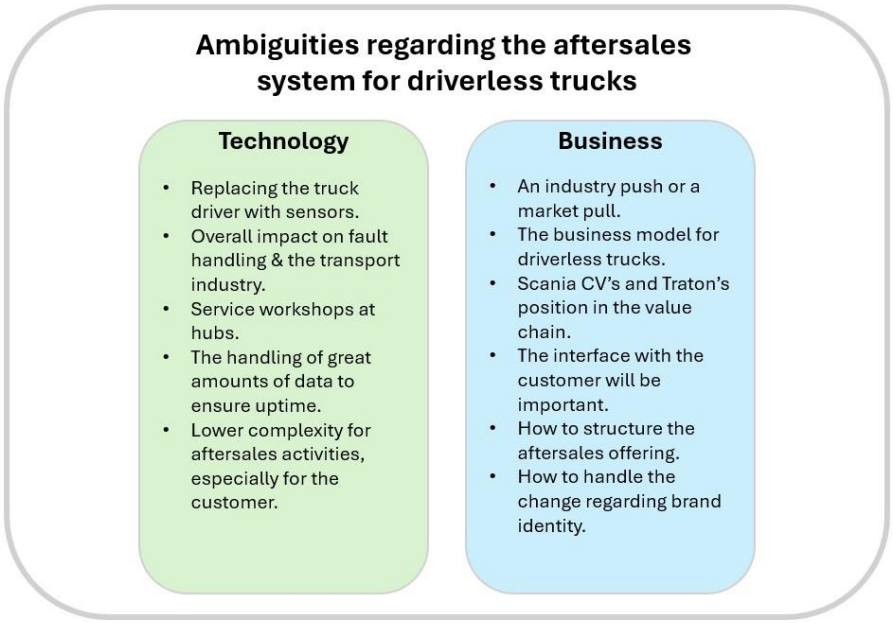



*Figure 9: The highlighted ambiguities regarding the Aftersales system for driverless trucks.*

*Technology-related ambiguity*

There is a high degree of ambiguity regarding how much driverless trucks will affect fault handling, as well as the business aspects of the aftersales offering. Ambiguity is defined as several interpretations of a specific state or future (Weick, 1995). This is shown by the findings from the empirical result of the thesis work, as there are many different interpretations of some aspects of the upcoming transition toward driverless trucks. One great example of where there is ambiguity is regarding replacing the truck driver with sensors. Some of the respondents saw it as a challenge since the truck driver supports the fault-handling process. Others argued that it was an opportunity as truck drivers could hinder the fault-handling process and that software could handle both fault-handling and safety aspects better than humans. There is also ambiguity regarding what impact driverless trucks will have on the transport system overall. Some respondents argue that the impact will be great as it implies an immense change for the whole transport system since driverless trucks can enable long haulage transport to be conducted on a 24/7 scale, which potentially can revise the complete transport system that we are used to today. One example is that there will be a need to establish hub-based workshops to make maintenance and service as efficient as possible to support the high utilization rate of the trucks.

On the other hand, other respondents argued that driverless trucks should only be added to the existing system and that the introduction should be conducted in small increments with minor implications for the road-based transport system. The respondents that argued for an incremental impact only raised the importance of educating service workshop personnel as they will need new competence to service the driverless truck software at the existing workshops. This problem is also raised by Porter & Heppelmann (2021), as they present the importance of adopting the shift

towards smarter products and services, which includes improving software handling capabilities. All of these ambiguities are difficult to handle but could be supported by one of the properties of sensemaking, "driven by plausibility rather than accuracy" (Helms Mills et al., 2010). As there are difficulties in finding the right and optimized answer to ambiguity, the involved actors should instead focus on what is most plausible to find an answer that makes enough sense to solve the ambiguities by reaching a consensus among the ideas related to the technological aspects of the aftersales system.

Porter & Heppelmann (2021) also stated that smart products are one of the largest trends among manufacturing firms and that software capabilities have become more important for a product´s capabilities. This is seen within the empirical data from the interviews, as some of the respondents state that more data will be needed to ensure uptime with driverless trucks. It will be complex to handle and analyze the great amounts of data. Many could argue that AI can be the solution to handle this in the context of fault handling; however, one of the respondents mentioned that many see AI as the solution for everything and that it will be a challenge to manage this. That more data is required to handle fault handling leads to driverless trucks being more complex for truck manufacturers, at least initially. After processes and methods have been established it might lead to simplified aftersales activities due to the enabling of the operator having more control of the vehicle as one respondent states. A respondent also states that it will primarily decrease the complexity for the customer as the manufacturer will have more responsibility for the driverless trucks. This is supported by existing literature, as Porter and Heppelmann (2021) presented that smart and connected products can help manufacturers integrate themselves better into the operation of the customer. Integration with customers will be important, as other respondents have also emphasized that the manufacturer of trucks will have more responsibility for fault handling, which will put demands on the manufacturer's capabilities to integrate themselves into the customer's operations.

*Business-related ambiguity*

There is also ambiguity regarding the business aspects of the driverless truck offering. One great example is regarding whether there is an industry push or a market pull for driverless trucks. According to Mittag et al. (2018), to make the servitization of a product profitable, there is a need for a balanced ratio of industry push and market pull. Otherwise, it is a great challenge to balance the technical capabilities of the organization with the needs of the customers. This is a challenge for Scania CV and Traton, as the majority of the respondents agree that the development of driverless trucks is mostly driven by the industry and not by the market. There is a minority that argues that there is a clear market pull. Those who argue for driverless trucks being a market pull also mentioned that it was more of an industry push previously but that it is both an industry push and a market pull now. At the same time, most of the respondents argued that the primary incentive for driverless truck development is the opportunity to create a more cost-effective transportation system and that customers always strive to have as low costs as possible. Thereby, it seems like there is a market pull for cheaper transportation alternatives and that driverless truck technology could be an enabler for this. However, the respondents disagree on whether driverless truck development itself is an industry push or a market pull. The ambiguity regarding who the customer is for driverless trucks is also shown in the empirical data. Some respondents argue that large-scale haulage companies in the US could be the first customers due to the size of the fleet and the country´s favorable legislation. Others are unsure who the customers are, and the fact that a majority of the respondents do not see driverless technology as a direct industry push is a challenge since it is important to collaborate with the customers and understand the needs to develop an appropriate business model and offering (Parida et al., 2014).

Another customer aspect that was raised by respondents was the interface towards the customer. It was presented as of great importance for the driverless truck offering; whoever succeeds with

this will be the market leader. This was also highlighted by Velamuri et al. (2013), who stated that it is important for organizations to integrate both their services and products into the same offering for their customers. This is also a field of ambiguity within the organization, as there are different ideas regarding how the interface towards the customer will be designed. However, there is a clear idea of how this should be achieved among the Autonomous departments, where high customer integration is promoted.

There is clearly a change that is occurring in the industry regarding the development of driverless trucks. In these kinds of environments, Velamuri et al. (2013) presented that business model innovation is important for organizations. This is a great challenge, as Engholm (2021) argues that it is unclear how the business model for driverless trucks will be defined. This can clearly be seen in the empirical data as well, as some respondents described the business model of driverless trucks as ambiguous. One of the respondents also stated that this is a challenge as the business model for trucks has not experienced any drastic changes in a long time, making it even more troublesome to start revising a well-rooted system. This is agreed upon by the literature as this affects the sensemaking process as retrospective is an important property to make sense (Weick, 1995). Therefore, it will probably be a challenge for Scania CV and Traton to solve the ambiguities regarding the business model for driverless trucks. Another aspect of the business model is the company's position in the value chain, which was also presented as ambiguous by the respondents. This can be related to the literature, as Wise & Baumgartner (1999) argued that servization gives an opportunity for organizations to expand horizontally in the value chain. However, if and how Scania CV and Traton should or should not expand is difficult to foresee as it would be a great challenge for them to start competing against their customers or suppliers.

The business case for driverless trucks has been presented as ambiguous. However, this also leads to that the structuring of the aftersales offering is ambiguous. Two of the respondents expressed that the business aspects of the aftersales offering are a challenge and that the organization does not yet have a plan for how to structure it. Another respondent raised the concern that aftersales will be more integrated into the driverless solution, which means that the involvement of the department in the development of driverless trucks probably needs to be high to ensure an appropriate solution where aftersales functions are more central. According to Monios & Bergvist (2020), truck manufacturers will have to change their business models radically as the trucks and services are offered in the same package. This challenge can be seen in the empirical data as well. Some of the respondents see a great increase in the need for preventive fault handling. Scania CV and Traton get revenue through service and repair when the customers return their trucks to the workshops.

If driverless trucks are offered as a service, Scania CV and Traton will instead decrease their revenue since money cannot be earned by maintenance activities. Uptime and preventive fault handling are important today to satisfy the customer. However, it will be more important to offer driverless trucks as uptime will be the most essential factor. This is directly supported by Velamuri et al. (2013), which highlights the importance of customer satisfaction when embracing PSS. The authors also raised the issue that the services should be more of a partnership-like relationship between the organization and the customers. This can be fulfilled by integrating fault handling with the base product, which, in this case, is the driverless trucks. The empirical data also showed another reason why uptime and preventive service will increase in importance: the driverless truck will have to be up and running to a higher degree due to its expenses and the opportunity of 24/7 operation.

Another aspect regarding ambiguity for the aftersales offering of driverless trucks is the importance of brand identity. A majority of the respondents shared the opinion that brand will be less important for driverless trucks, as vehicles will be seen more as a commodity. Today, Scania CV’s brand is tightly connected to fuel consumption and the driving experience for truck drivers.

Instead, with driverless trucks, the total cost of ownership and the uptime will be the most important aspects. The change regarding brand identity will make it harder to differentiate Traton's brands. Therefore, the question of whether there is a need for brands has been raised. This affects aftersales as their activities are highly connected with brands due to the customers' relationship with the branded workshops that they frequently visit. Also, there are great challenges with not having brands today, as presented in the empirical data. One is that the customer interactions are handled through the brands, and some of the customers are not familiar with Traton, thereby making it difficult for the organization to restructure and build relationships through Traton itself.

Another challenge is the cultural aspects of the brands, as they differ, and there is a challenge in merging different cultures. The last challenge is presented by Martinez et al. (2010), as they argue that organizations must review their internal environment, for example, culture, when transforming their offering to a PSS structure. Brand identity is a great upcoming problem for Scania CV and Traton, which is highly related to sensemaking, as identity construction is one of the included properties (Helms Mills et al., 2010). There is a need for Scania CV and Traton to understand their new identity regarding brands and how or if they should create a differentiation between the brands within Traton. It is also important that all the involved departments understand what their identity is in this change so that everyone works in the same direction. The strong connection between brands and aftersales makes it crucial to solve to the brand question to succeed with the aftersales offering of driverless trucks.

*Different visions between departments and hierarchal levels*

The above-presented insights conclude that there is proof of ambiguity regarding the aftersales system for driverless trucks in terms of both technical and business-related aspects and that various visions are established within the organization of Scania CV and Traton. There are most likely multiple underlying reasons for the ambiguity, but a potential cause is the fast-paced development of the driverless truck field, making it more of a radical innovation with an uncertain future.

One of the hardest parts of managing technology development is handling shifts between radical and incremental innovation (Benner & Tushman, 2003). According to Dosi (1982), discontinuous innovations lead to shifts in technical trajectories since the technological paradigm gets reshaped by the discontinuities. Since the boundaries for possible technological directions are affected by discontinuous elements of innovation, it is reasonable to assume that ambiguity arises if the scene for technological development is constantly changing. In the case of driverless truck development, multiple respondents have emphasized how the perception of both technological and business-related aspects of the driverless truck offering has changed during the past years. A possible cause for the ambiguity within Scania CV in regards to driverless truck development is the fact that there is no clear high-level vision for driverless trucks that is communicated to all departments of Scania CV and Traton, except to those who are responsible for driverless truck development. The primary reason for this is that driverless truck development stands for a rather small part of Scania CV's R&D budget, but it still creates disconnections and makes it hard to establish a general understanding of the current development trajectory among the whole organization.

As presented by Lewis & Clark (2020), sub-units within larger organizations tend to develop individual visions that sway away from the more abstract and high-level one of the whole organization, which, with the combination of a fast-changing technological field, can lead to quite drastic differences in perception between departments who are responsible for working with the specific technological field, compared to departments with other focus areas. The empirical data shows that different visions and expected challenges between departments are present. It is obvious that some departments have a clear vision regarding the first customer and the customer

needs for driverless trucks in a hub-to-hub context, while others are lacking one completely. The departments that are developing driverless trucks today, the Autonomous – R&D and Autonomous – Business departments, see a clear market pull and whom the first customer is as opposed to the other departments who only see the development of driverless trucks as an industry push and no clear customer needs. It will be critical for the Autonomous departments to communicate the needs of the customer to other parts of the organization to construct shared visions within the organization. However, there are also shared visions and expectations between the departments that have been presented in the empirical data.

Different visions can exist on different hierarchal levels, and there is a need for collaboration to avoid this problem (Hodgkinson, 2022). The need for collaboration is also needed in sensemaking, as social interaction is where the involved individuals make interpretations (Helms Mills et al., 2010). The authors also present that it is important for the organization to create an environment where sensemaking is supported and that cues are an important part of the involved individuals to make sense of and create new interpretations. Therefore, it is important for the organization to enable collaboration to create social interaction as well as include individuals from different hierarchical levels to gather cues from different perspectives that support the sensemaking process, which can result in more collective visions. The empirical data shows that different visions between hierarchal levels are an issue within Scania CV and Traton. One example is that there are some differences within the Aftersales - R&D department between the upper and lower/non-managers regarding the opportunities to replace the truck driver with sensors. Where the lower managers see more opportunities to replace the truck driver, and the upper managers see more challenges. This is an interesting and differing vision as the truck driver today is a central part of fault handling, which is primarily the focus of the Aftersales – R&D department. The empirical data also show a lot of shared visions and expected challenges for the fault handling and business aspects of the aftersales offering between hierarchical levels within departments.

## *5.2 Implementation of Lewins three-step model*

As presented in the Frame of Reference, Lewin's change management model, or three-step model, is a framework for managing organizational change (Burnes, 2004). The model consists of three stages: unfreeze, change, and refreeze, and can be implemented and iterated to revise organizational processes or structures that are insufficient (Burnes, 2004). In this section, Lewin's model will be utilized to answer research question 2 - *How can guidelines be formulated to address the challenges identified in the development of the aftersales systems for driverless trucks based on a comprehensive understanding of the current status and envisioned future of the system, with the help of change management theory, specifically utilizing Lewin's three-step model*?

This will be conducted in three steps:

1. Unfreeze: Analysis of the current organizational practices toward developing the aftersales system for driverless trucks, based on theory from the Frame of Reference and empirical data from the Results chapter.
2. Change: Proposals for how the development could be conducted to address the challenges identified in the empirical data and the previous step of the analysis.
3. Refreeze: Suggestions of indicators for when a desirable outcome has been achieved.

Figure 10 illustrates the three stages of Lewin's change management model and provides summaries of each of the following subchapters.

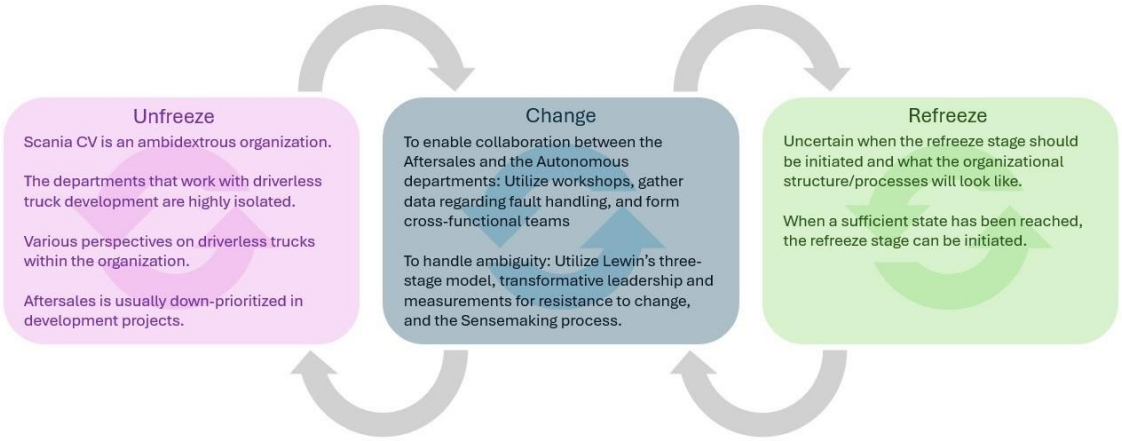



*Figure 10: Illustration of Lewin's three-stage model for Scania CV and Traton.*

### 5.2.1 Unfreeze

Ambidextrous organizations designate product development into different subunits to conduct incremental and radical innovation simultaneously effectively and efficiently (O'Reilly III & Tushman, 2004). To make an ambidextrous organization successful, it is important to balance separation and collaboration among the subunits to allow for both resource and knowledge sharing and agility and flexibility (Patanakul et al., 2012). Also, there is a need for clear vision communication across the subunits to support a shared understanding of the organizational goals and how they can be achieved (O'Reilly III & Tushman, 2004).

Scania CV utilizes an ambidextrous organizational strategy for its R&D and business-related activities, as there are various departments with different focus areas, both in terms of technological or business-related scope and also regarding the innovative character of its focus areas. According to multiple respondents, the departments that are working with driverless truck-related development are isolated from the remaining subunits of the organization, allowing them to adopt more agile processes and be more flexible, which is desired to manage the uncertain nature of driverless truck technology. As a result, the collaboration between the Autonomous departments and remaining subunits at Scania CV and Traton is limited since it is hard to establish a well-functioning collaboration between subgroups with processes and cultures that are tailored for incremental and radical development, respectively.

For instance, one of the inhibiting factors for collaboration between the Aftersales and Autonomous departments is the drastic differences between the subunits' internal processes and cultures. According to the respondents, Scania CV is implementing more agile processes overall to decrease organizational rigidity and become more flexible. However, there are major differences among the organization's subunits as their focus areas require different kinds of processes. The biggest difference between the aftersales and autonomous departments is that the autonomous subunits, working with driverless truck technology, have a strong bottom-up culture and put heavy emphasis on vision implementation to support future work, while aftersales do not seem to have a clear direction for future development and instead focuses on solving short term issues without really having a long-term vision for driverless trucks. As emphasized by (Gratton, 1996), one of the biggest barriers to unified vision implementation is organizational structure and culture, and integrating two subunits with radically different cultures and ways of operating can thereby be a challenge, as a common vision must be formulated in a way that compels both of the cultures and can be digested by each subunit. As discussed in the previous section, there are many

ambiguities regarding driverless truck development in general, but specifically for the aftersales-related aspects of the system. To cater to these ambiguities, the aftersales and autonomous departments must collaborate to make sure that knowledge is shared and that a sufficient aftersales system is designed.

Also, the outlook on driverless truck development is heavily colored by personal opinions. One of the respondents expressed that people seem to neglect facts and rather base assumptions on their own perception regarding the future of driverless trucks. This might be a result of the immensely ambiguous character of the driverless truck field, leaving space for speculation, which ultimately results in various perceptions of what the future will look like. For instance, older visions that have been shown to be infeasible might still be kept close to heart by some practitioners, while others move on to acquire new ones. This is primarily illustrated by the various thoughts regarding the time frame for driverless trucks regarding hub-to-hub applications. More specifically, the respondents who work with the development of driverless trucks have clear visions of when the trucks will be available for commercial operation, while others who have no clear connection to the driverless truck departments seem to base their assumptions on feelings and general industry climate.

As mentioned by Lewis & Clark (2020), the evolvement of diverse visions among subunits of large organizations is a common phenomenon as the subunits get disconnected from each other. Also, shifts in technological trajectories and paradigms lead to new assumptions and ideas about the future of a technological field, resulting in a new vision (Dosi, 1982). These perspectives are shared by two of the respondents as well, as they express that Scania CV can be seen as a group of "smaller companies" that collaborate with each other and that disconnections among them can lead to separate vision development. Since driverless truck development is only conducted by isolated departments at this stage, other subgroups of the organization might simply lose track of the strides that are being made in the driverless truck field, what is regarded as feasible or not, and how it affects the vision for future development.

Furthermore, multiple respondents mentioned that resources and organizational interest usually is directed toward pure product performance and specifications at the expense of aftersales-related development. Even though the empirical data indicated that there is a well-established understanding of the importance of aftersales activities for trucks in general as well as specifically for driverless trucks, aftersales aspects often get taken into consideration late in development projects and are regarded as less interesting. One of the respondents mentioned that this seems to be the standard scenario for multiple industries, as aftersales have always been a second-order priority for truck and bus manufacturers, and pure product performance has been the core element in the offering.

However, many of the respondents emphasized the importance of integrating aftersales earlier in the processes to support the servitization elements of Scania CVs' offering as it is a great prospect for revenue. Also, collaboration among subunits is functioning well overall as there are greater incentives for aftersales integration in Scania CV's current product range. In the case of driverless trucks, respondents from the aftersales department at Scnaia CV mentioned how they have tried to be involved in the driverless truck development but that the product was too immature at that stage, leading to insufficient involvement of aftersales resources. Yet, as stated in (Porter & Heppelmann, 2021), servitization and aftersales activities play an important role in business offerings of smart and connected product systems, and various respondents highlighted the need for including aftersales in development projects for driverless trucks relatively soon to make sure that the right decisions are made.

One of the managers from the autonomous departments mentioned that aftersales must start to gather knowledge of driverless truck technology to understand how their processes and methods

can be adapted for driverless trucks and that it is time to integrate the department when there is a well-defined driverless truck concept. One area where collaboration will be needed is how fault handling should be managed, as multiple functions of the organization will be involved in solving potential errors in the driverless truck platform. However, one of the respondents mentioned that there is a general interest among Scania CVs' various subunits in how aftersales function and how they work, as previous workshops to share knowledge of aftersales operations have been appreciated. This suggests that the inclusion of aftersales in driverless truck development can be fruitful as long as there are incentives and clear objectives for the integration. Therefore, to create a foundation for cross-functional collaboration for the aftersales aspects of the driverless truck offering, implementation strategies for common vision establishment must be utilized to limit the influence of individual perception and behavioral barriers.

### 5.2.2 Change

According to various respondents, cross-functional collaboration among the autonomous and aftersales departments must be initiated to address the emerging need for aftersales development for driverless trucks. This also applies to business-related aspects of the driverless truck offering, as there is a need for thorough technological competence in driverless truck development to make sure that sufficient business models are created. As for specific collaboration between the aftersales and autonomous departments, one of the respondents highlighted that driverless trucks must be out on the roads in order for aftersales to collect real-life data to be able to truly understand how they should streamline or adapt their processes for driverless fault management properly. However, in order to start creating knowledge of driverless technology to make future collaboration as seamless as possible, many of the respondents argued that collaboration should be initiated as soon as there is a well-defined concept that can be evaluated by the different departments and that preparation for collaboration should be started rather soon.

One of the most critical aspects and apprehension among the respondents is that the aftersales department must consider driverless trucks as a completely new challenge and not rely on old processes to make sure that a truly innovative output is generated if needed. This might be a challenge as the current culture of Scania CVs aftersales department is centered around short-term development, which can induce issues when shifting focus toward a novel development area. However, as one of the respondents from the aftersales department pointed out, it is important to take one step at a time, try to analyze what the future holds, and make incremental changes early to avoid having to make great heaps to catch up later on. Also, when starting to collaborate closely with the autonomous department, a subgroup from the aftersales department will most likely be assigned the task of detecting and analyzing future requirements for an aftersales system optimized for driverless trucks, allowing the subgroup to develop their own processes to support radical or uncertain development, just as described by (Patanakul et al., 2012).

Another important aspect to bear in mind when establishing cross-functional collaboration between the autonomous and aftersales departments is that the flexibility of the autonomous subunits might have to be reduced to support conditions for knowledge and resource exchange as external dependencies might need to be increased. As expressed by Mikalsen et al. (2019 and Patanakul et al. (2012), external dependencies to other subunits can limit the creative and agile capabilities of an autonomous team and thereby hinder their ability to function as effectively as possible in radical innovation contexts.

However, one of the managers from the autonomous departments mentioned that focus always will shift between allowing explorative freedom and “framing” to focus on specific tasks, similar to a sinus wave where the degree of current uncertainty or ambiguity oscillates. Since the collaboration with the aftersales department should not be initiated before a driverless truck

concept has been finalized, the collaboration will take place when ambiguity related to the driverless technology has already been managed, and the product has reached a certain maturity level. This might reduce the need for internal flexibility and allow for better collaboration conditions, as the next step in the development will be to share resources and knowledge to better understand how the aftersales system should be structured. Less need for internal flexibility and a higher degree of explorative freedom might also be a situation where increased inter-organizational collaboration can be a good addition, establishing an opportunity to gather insights from other industries or partners with appropriate proximities, as it can support future development and understating of how the business aspects should be structured (Knoben & Oerlemans, 2006).

Yet, the most important factor to bear in mind to actually conduct organizational change, according to (Burnes, 2020), is to have an iterative mindset and test different development strategies and organizational structures to find one that has a desirable outcome. Since the driverless truck field is embossed by ambiguity, development and innovation strategies will need to be revised in the case of unforeseen paradigm and technological trajectory shifts to support the development of the aftersales system for driverless trucks, meaning that management should be open towards trying different strategies and evaluate their methods continuously to find a sufficient strategy for aftersales development.

More precisely, transformative leadership is a fitting leadership style to use for Lewin´s three-step model. Therefore, if Scania CV adopts this change management model, it should also adopt a transformative leadership approach to handle the change towards driverless trucks. As presented by Hussain et al. (2018), the leadership approach includes collaborating, sharing knowledge, and involving others in the decision-making process. This would be fitting as there are different visions within the organization regarding fault handling and the aftersales offering. Therefore, there is a need to share knowledge between departments and hierarchal levels within the organization in order to collaborate more.

Involving employees in the decision-making process could also support finding where there is resistance toward the change to driverless trucks. Scania CV and Traton should also revise their perspective of resistance to change. The organization could start to utilize the two measurements of resistance to change presented by Cinite & Duxbury (2018). The two measurements, “Commitment to organizational change” and “Voicing concerns about change”, support the organization to create an environment that is supportive and has low job stress levels, as well as an environment where the employees want what’s best for the organization and therefore provides improvements regarding the change. This is supported by the transformative leadership approach, as it motivates the employees to take the initiative (Michaelis et al., 2010). Adopting Lewin’s three-step model together with a transformative leadership approach and utilizing the two measurements for resistance to change could support Scania CV and Traton in creating an organization with employees that can handle the change towards driverless trucks.

Scania CV and Traton should also consider the sensemaking process created by Weick (1995) to handle the current state of the organization regarding the development of the aftersales system for driverless trucks. This development is affected by a high degree of ambiguity, and the sensemaking process is an appropriate approach for handling ambiguity (Weick et al., 2005). However, Scania CV and Traton need to address the challenges of implementing and utilizing sensemaking, too. One of the challenges is the politics of sensemaking, which could lead to upper managers prioritizing their involvement in sensemaking (Maitlis & Sonenshein, 2010). The other challenge is that sensemaking in itself is ambiguous and can be difficult to implement and utilize (Sandberg & Tsoukas, 2015).

Due to these challenges, the properties of sensemaking created by Weick (Helms Mills et al., 2010) have been reconstructed in an attempt to clarify how Scania CV and Traton could use sensemaking to solve the ambiguity regarding the aftersales system for driverless trucks.

Firstly, **grounded in identity construction**. Scania CV and Traton need to figure out their new identity regarding brands. What will their new brand identity be for driverless trucks, and how can they create differentiation between the brands within Traton? Every department and individual involved needs to understand their identity in this change.

Secondly, **retrospective**, where the involved departments and individuals should use previous experiences to create an understanding of the future aftersales system for driverless trucks. A good example is the existing increasing need for uptime and proactive fault handling; how can knowledge of today's system and previous systems solve the ambiguities of the future system?

Thirdly, **enactive of sensible environments**. Scania CV and Traton should create an organizational environment where the employees are encouraged to seek an understanding of the aftersales service system for driverless trucks, and that enables the other properties within sensemaking.

Fourthly, **social**. As social interaction creates understanding, collaborations should exist between departments and different hierarchal levels so that all involved individuals have a chance to participate in the sensemaking process. This is important to avoid negative consequences regarding the politics of sensemaking.

Fifthly, **ongoing**. Scania CV and Traton need to understand that sensemaking is a constant process. Therefore, iterations of the properties are important, and knowledge to solve the ambiguities should be constantly sought and updated.

Sixth, **focused on and by extracted cues**, all the involved individuals should seek and construct cues to better understand the change. It is important that all departments and hierarchal levels collaborate to create cues as different cues make sense for different individuals.

Seventh, **driven by plausibility rather than accuracy**, finding the optimized ideas is not supported by sensemaking. Therefore, ideas that are plausible should be the focus, and thereafter, actions to test these ideas.

All of these guidelines support each other as the workshops utilized for knowledge sharing and cross-functional teams support the sensemaking process as they can enable social interaction and the creation of cues between departments. The iterative nature of Lewin's three-step model enables the ongoing sensemaking process. The transformative leadership style argues for involving employees, which enables increased collaboration. The resistance to change measurements creates an organizational environment where employees are encouraged to raise concerns and ideas, which assists the sensemaking process as it is in need of an environment that encourages knowledge seeking.

### 5.2.3 Refreeze

Refreeze, the last stage of Lewin's three-step model, cannot be covered in this thesis project, as the suggested change has yet to happen. As mentioned, the model is iterative (Burnes, 2020), so one recommendation could be to reach this stage as fast as possible by implementing parts of the suggested guidelines in stage two of the model. Thereafter, evaluate the changes in this stage and reinforce the changes if the results are positive.

## 5.3 Sustainability aspects

The greatest driving force regarding driverless vehicles is to reduce the TCO as you replace the driver, which itself is a great cost (Engholm, 2021). The empirical data confirms this, as many of the respondents stated that the greatest driver for the development of driverless trucks from their perspective is the TCO. It was also presented during the interviews that the driver is approximately one-third of the total cost. Therefore, there are a lot of cost savings that could be done by replacing the driver with an autonomous driver. The empirical data also showed that another benefit of replacing the driver is that haulage companies do not have to consider the existing time restriction for human drivers. Therefore, driverless trucks can idealistically enable 24/7 operation on the road. Engholm (2021) also stated that further research is required regarding the sustainability and societal effects of driverless trucks. During the interviews, the environmental drivers and impacts were rarely mentioned, which is interesting as the environment is an important perspective for almost all organizations.

One thing that was discussed during an interview regarding the environment was how modularization is great for aftersales as it affects serviceability positively, but it can also affect the environment negatively. Due to that, you can change whole modules if one component within the module has a fault. However, this does not specifically pertain to driverless trucks but to trucks in general. Social aspects were mentioned, and two of them were presented in the empirical data. The first one is regarding safety, as some respondents argue that driverless trucks could be safer compared to human truck drivers. They viewed safety as a driving force for the development of driverless trucks. The other one was regarding legislation. This is an enabler as well as a barrier for driverless trucks. Therefore, the ones responsible for legislation regarding transport must consider the development of driverless trucks and how it affects the development.

## 5.4 Method discussion

To cater to the downsides of qualitative research, the research method of the thesis project has been inspired by already proven methods. Discussion with the project supervisor was also a central part of assessing the credibility of the research method to make it as trustworthy as possible. Furthermore, continuous work has been done to ground the assertions and findings with evidence gathered through the data collection methods. The material from the semi-structured interviews has been evaluated through iterations of reading, discussion, and analysis to avoid fragment picking and ensure a full understanding of the context of the findings. The findings have also been thoroughly discussed with the project supervisor to further minimize misunderstandings of its importance and context. In addition, the respondents were informed of the conclusions drawn from the analysis to allow for revision if their statements had been misinterpreted. Also, the project's result is more of an abstract manner and should not be viewed as an implementation strategy for the development of aftersales activities for driverless trucks. Instead, the findings should be considered as general guidelines or suggestions for supporting the development of the aftersales offering for driverless trucks.

However, to establish a more fair representation of reality, more interviews could have been conducted, both in terms of the number of respondents from each department as well as involving respondents from other departments that will be involved further in the future of driverless truck development. This is especially relevant for interview round two as a greater quantity of respondents probably would generate more nuanced data, which could have supported the quality of the results. However, due to the time restriction of a thesis project, there was simply not enough time to expand the second interview phase. This is a con of qualitative research and grounded theory, as it is a time-consuming process (Bryman & Bell, 2017).

In order to assess qualitative research, the following criteria should be taken into consideration: credibility, validity, reliability, and justification of the method (Bryman & Bell, 2017). This has

been achieved as the research process was pursued with care and as consistently as possible. The respondents were also contacted after the empirical data was formulated to ensure that the respondents agreed with the material published to achieve credibility for the thesis project. The research setting for this thesis project was very specific; however, to ensure validity, the findings of the research are general and probably applicable in other settings where there are high degrees of ambiguity regarding technological development and where collaboration is needed. The method description and research process of the thesis project have been clearly presented to avoid affecting the reliability of the work. No personal agendas regarding the research topic were apparent from the start either, making sure that the analysis was affected by bias or subjective opinion as little as possible to support the criteria of "justification of method". However, it is difficult to avoid personal opinions inflence the work completely. An example is regarding the coding of the interviews, as the codes could probably have been affected by personal preferences. However, the implications and consequences of letting personal opinions affect the research have been taken into consideration and, thereby, been limited as much as possible.

On a final note, to ensure a report with sufficient academic language, the AI tool Grammarly was used to improve sentence structure and conduct spell-checks on the material. The report has been revisited after the utilization of the AI tool to ensure that no misalignments in the presented information arose as a result of the spell check and reformulations. The authors of this thesis therby take full responsibility for the published material.

# 6 CONCLUSIONS & FUTURE WORK

*In this chapter, conclusions and recommendations on more detailed solutions and/or future work in this field are presented.*

## 6.1 Conclusions

The current vision among the autonomous departments is that driverless trucks should be seen as an addition to the existing road transport system as it will not revolutionize the industry, at least at the beginning of its era. From a technical perspective, there will be minor changes that affect the aftersales system; for example, the existing service network with workshops will probably be utilized, and the driverless trucks will have many similarities to manually operated ones. However, there are great upcoming challenges that need to be addressed, such as how to replace the truck driver's role and sensing capabilities in the fault-handling process with an autonomous driver or robot.

The assumed effects of driverless trucks were regarded to be more radical a couple of years ago. Now, they are seen as being more incremental. One reason for the revised outlook on driverless trucks' industry impact might be that ambiguity has decreased, but having a more incremental approach toward the development might also suit Scania CV as their development processes and organizational culture have always been directed toward incremental innovation.

There is a common belief among the respondents that uptime and proactive fault handling will be increasingly important for driverless trucks. This could be a result of the current focus on optimizing uptime and proactive fault management within the organization, as it is highly relevant for conventional trucks as well and a primary objective in Scania CVs product development. This makes it easy to transition the vision to driverless trucks as well, as it already is a priority for multiple departments. The same goes for modularization and AI, as the themes have high priority within the organization.

However, when it comes to nuances of the driverless truck platform and additional aspects that have no clear correlation to the current focus of the organization's departments, it is clear that ambiguity is apparent and that people's perceptions of the future start to diverge. A homogenous picture is only apparent among the respondents who belong to the autonomous departments, which primarily can be explained as they are responsible for driverless truck development and work with the technology.

Scania CV has a deep-rooted culture of prioritizing customer value instead of profit ratios and revenue potential in its product offerings. Yet, there are various perceptions of the amount of market pull for driverless trucks among the respondents, making it troublesome to understand what the customer wants if they cannot be targeted. Even though there is a common understanding of the potential of decreasing the TCO with driverless trucks and that customers want as cost-efficient transport solutions as possible, there are still contradictions within the organization regarding whether driverless truck development has a clear market pull or not, something that might create challenges when involving multiple departments for cross-functional development.

The strongest driving force for the development of driverless trucks is economic efficiency, as cost reductions can be made by removing the driver. Incentives for the two other sustainability aspects, environmental and social sustainability, are weaker and more uncertain, which is emphasized by the literature and also by the empirical data, as the respondents primarily focused on the economic possibilities of driverless trucks.

Currently, there is limited collaboration between the Aftersales and Autonomous departments regarding aftersales system development. However, the respondents agree that it will soon be time to integrate and start creating a foundation for cross-functional development. Integration of the

aftersales department should be initiated when there is a proven concept for the technology-related aspects of the driverless truck platform, as it is the primary barrier that will allow for collaboration. The business-related aspects of the offering are too ambiguous yet and need more time before they can be defined.

The transition to driverless trucks necessitates prioritizing aftersales aspects due to the servitized nature of the offering. To enhance collaboration, workshops can be used for knowledge sharing, leveraging Aftersales - R&D's expertise in fault handling processes. Cross-functional teams comprising technology and business experts are essential for developing a comprehensive aftersales offering. Scania CV and Traton can manage the complex and ambiguous transition using Lewin's three-stage model together with transformative leadership and the resistance to change measurements, as well as sensemaking to foster innovative solutions and effective change management. The proposed guidelines desirably complement each other, meaning challenges regarding collaboration and managing ambiguity can be handled by combining them. All of them support the involvement of all affected departments and actors in solving the ambiguities regarding the aftersales service system for driverless trucks.

Finally, it is important to remember that driverless truck development only stands for a small part of Scania CV's focus today, as most of the organization's resources are allocated to electrification and driveline development. If autonomy had represented a larger part of R&D's development strategy, more resources would probably have been devoted to streamlining the vision distribution and collaboration efforts among all relevant departments for the development of a driverless technology solution. The receptivity to autonomous visions would certainly have increased as well if the overall feeling among the Scania CV's departments had been that autonomy is everyone's main focus.

## 6.2 Future work

As mentioned, this thesis project cannot cover stage three of Lewin's change model, Refreeze, due to time restrictions. Therefore, it is proposed that future research be conducted on the transition towards driverless vehicles and how to solve the ambiguity regarding fault handling and the aftersales offering for driverless trucks. There is a need to evaluate the change and thereafter enforce the utilization of the proposed recommendations if the results are positive.

The choice of change management model could be further investigated. Due to the time restriction, there was no in-depth analysis and comparison of different change management models. It would be interesting to compare Lwein's three-step model with others to evaluate which one is the most appropriate for this case.

There is a need to further investigate when there is time to start the development of the aftersales offering and when to involve the Aftersales – R&D department in the development of driverless trucks. As this thesis project was conducted under a tight time constraint, it would be beneficial to conduct more interviews from the chosen departments in this thesis project as well as other departments. Future work could also include doing other types of research, both qualitative (e.g. workshops) and quantitative (e.g. surveys).

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# APPENDIX A: INTERVIEW GUIDE ROUND ONE

**1. Introduction – 2min**

We will map out visions of an aftersales service system for driverless vehicles in a H2H context and then construct road maps.

We are going to record the interview.

We will check the transcription and quotes with you.

**2. Who are you? – 3 min**

Can you briefly tell us who you are? (Role at MAN, background/previous positions)

Who do you collaborate with at MAN?

What knowledge do you have about aftersales?

**3. Driverless vehicles - 5 min**

Why is MAN developing driverless vehicles & what are the greatest driving forces?

**4. Aftersales service system - 5 min**

How do you think the transition to driverless vehicles will affect the aftermarket/aftersales?

What demands do you think the aftermarket/aftersales will place on the development of driverless vehicles?

**5. The vision of an aftersales system for driverless vehicles – 20 min**

What is your vision for the aftersales system for driverless vehicles?

A personal or the organization's vision?

How should driverless transport solutions be offered to the customer (DaaS vs. TaaS / Value chain)?

Requirements & needs today and in the future?

Which trends do you think will affect the development of this? (legislation, environmental requirements, technology)

**6. Visions - 20 min**

Do you work with visions of the future?

Why and how/who creates the visions?

How do you work to transition from the present to the desired future? / How are the visions used?

How is it communicated? (text, image, roadmap, etc.)

Is it a disadvantage of setting up concrete/clear visions of a distant future (Concept of autonomous mining vehicles)?

(Who contributes with knowledge about the current situation? How is this validated?)

How do you handle uncertainties in the 'vision development' - e.g., radical innovation vs. incremental

**7. The end - 5 min**

Can Scania's modular system facilitate / inhibit the implementation of driverless vehicles?

Is there anything that you would like to add?

Why is MAN developing driverless vehicles & what are the strongest driving forces?

**Triggers**

1. Anything we missed regarding [a specific area]?
2. Will the integration between the truck and add-ons need to be improved?
3. Delimitation/compilation of data that is transferred remotely. How much should be sent? Pro and post error/fault.

# APPENDIX B: INTERVIEW GUIDE ROUND TWO

**1. Introduction – 2min**

We are going to map out visions of an aftermarket system for autonomous vehicles in an H2H context and then construct roadmaps.

We will record the interview.

We will check the transcription and quotes with you.

**2. Who are you? – 3 min**

Can you briefly introduce yourself? (Role at Scania, background/previous positions)

Who/which teams do you collaborate with at Scania? (EA, YS, ZS)

What is your knowledge of the aftermarket system?

**3. Autonomous vehicles - 5 min**

Why is Scania developing autonomous vehicles & what are the strongest driving forces?

What is the customer demand for autonomous vehicles?

What impact will autonomy have on the industry?

**4. Vision of the aftermarket for autonomous vehicles – 20 min**

How will the requirements change for the aftermarket of autonomous vehicles?

Is it difficult to predict how the aftermarket will need to adapt?

- Does the aftermarket have a lower priority?

How will the complexity of the aftermarket system change?

- The driver's role in case of faults?

How will the workshop network be affected?

- Workshops at hubs?

How should autonomous transport solutions be offered to the customer?

- Servitization
- Integration with the customer
- Brand Identity (TRATON) & commodity

What key trends do you think will affect the development of this? (legislation, environmental requirements, technology)

Can Scania's modular system facilitate / counteract the aftermarket for autonomous vehicles?

**6. Visions & working towards the future - 10 min**

Do you work with visions of the future?

Is it easy to start working with new ideas (Bottom-up approach)?

How do you work to move from the present to the desired future? / How are the visions used?

How do you handle high **uncertainties** - e.g., radical innovation vs incremental

**7. Conclusion - 5 min**

Is there anything that we have missed or that you would like to add?

What timeframe do you see for autonomy?

**Triggers**

1. Have we missed anything [about specific area]?
2. Limitation of data - should not be too much (data before & after faults)
3. How can the aftermarket be more efficient? Not just focusing on more efficient transportation
4. Challenges with legislation and public perception

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